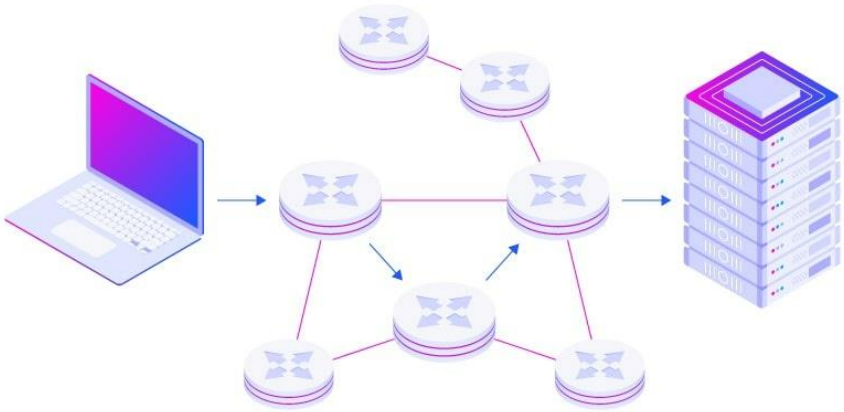
U.PORTO

FEUP FACULDADE DE ENGENHARIA
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Relatório

RCOM



3LEIC04

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Resumo

Este relatório foi realizado no âmbito da cadeira Redes de Computadores, do primeiro semestre do terceiro ano da Licenciatura em Engenharia Informática e Computação (LEIC). Consiste essencialmente na implementação de uma aplicação de *download* de ficheiros, numa primeira parte, bem como a configuração de uma rede de computadores, tanto física, ao colocar os cabos de rede nas posições corretas, como virtualmente através do uso de comandos no terminal.

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Introdução

Sendo o objetivo do trabalho laboratorial a aplicação prática dos conhecimentos adquiridos sobre o desenvolvimento de um cliente FTP (*File Transfer Protocol*) e a configuração de uma rede de computadores, a elaboração deste relatório focar-se-á nessas duas componentes. Deste modo, o documento terá a seguinte estrutura:

- **Aplicação de *download* –** Baseada no protocolo FTP, explicaremos brevemente a nossa implementação e apresentaremos os resultados apresentados pela mesma.
- **Configuração da rede –** Realizada durante as aulas práticas e descrita pelos guiões, descrevemo-la passo a passo, prosseguindo de seguida à sua análise.
- **Conclusões –** Apresentamos a nossa opinião em relação à análise do projeto como um todo
- **Anexo –** Documentos relevantes à interpretação do relatório, nomeadamente o código desenvolvido para a aplicação de *download*, os comandos utilizados para a configuração da rede e os *logs* registados ao longo das aulas laboratoriais.

De notar que realizámos a configuração da rede na bancada 2.

Parte 1 - Aplicação de *download*

O objetivo deste trabalho consiste em desenvolver uma aplicação que executa o download de qualquer tipo de ficheiros a partir de um servidor com o protocolo FTP(file transfer protocol) e com ligações TCP(Transmission Control Protocol) a partir de sockets. A aplicação aceita como argumento um link com a seguinte configuração **ftp://[<user>:<password>@]<host>/<url-path>**

Esta aplicação de download está divida em três partes: a Main, o Parsing dos argumentos e a Connection.

Arquitetura

Parsing do URL

Primeiramente realiza-se o parsing do URL que é fornecido como argumento na linha de comandos. A função parse_args() realiza o processamento da informação obtida através do URL, começando por armazená-la numa instância da struct arguments. Esta estrutura contém diversos campos tais como:

- User
- Password
- Host
- Path
- Nome do ficheiro
- Nome do host
- IP

Para preencher os últimos três campos são utilizadas duas funções get_ip() e get_filename(). A primeira função permite obter o IP e o nome do host, através de uma função que já nos é fornecida

denominada gethostbyname(), e a segunda função extrai o nome do ficheiro que se pretende descarregar e que corresponde ao último elemento do path. Desta forma quando o programa terminar, o ficheiro já tem um nome atribuído.

Connection

No que diz respeito a esta camada, de uma forma geral lida com o controlo e a transferência de dados. Tudo se inicia com a função connection_to_server() cujo código nos foi fornecido, que no fundo cria um socket de controlo com o IP e a porta recebidos nos seus argumentos.

A função write_to_server() escreve um comando para o servidor através do socket de controlo.

A função login() processa as credenciais do utilizador (user e password) incluídas no path do URL e coloca-as no struct arguments.

A função read_response() usa getline() para esvaziar o buffer que contém a resposta do servidor e tem um código de verificação caso o programa dê erro , para terminar em vez de ficar bloqueado. Na função enter_passive_mode() ocorre o envio do comando pasv , que é processado até ser obtido o novo endereço IP e a nova porta que é retornada no fim da função.

A função retrieve_file() recebe o ficheiro e após a sua leitura retorna o seu tamanho.

A função download () cria, finalmente, o ficheiro a ser transferido para a máquina, e preenche-o com os dados que recebe dos seus argumentos. É efetuada a leitura do socket de dados ao mesmo tempo que esses mesmos dados são escritos para o novo ficheiro, para no fim serem lidos pela função read_response() . No fim, por precaução é feito o fecho do ficheiro.

A função end_connection() processa o envio do comando quit , ao terminar a ligação tanto do socket de controlo como do socket de dados, consistindo numa uma boa prática.

Main

Neste módulo, todas as funções descritas anteriormente se combinam para efetuar o download do ficheiro.

Inicialmente ocorre a verificação da passagem de apenas um argumento na linha de comandos, na medida em que se isto não ocorrer o programa termina com uma mensagem de erro. Seguidamente, inicializa-se todos os campos da struct arguments e verificamos se a função parse_args() não tem erros, e neste caso é imprimida uma mensagem com os campos da struct devidamente preenchidos pela função.

Assim, entramos na parte Connection ao criar um struct TCP com os seguintes campos:

- Socket de controlo
- Socket de dados
- Novo IP
- Nova porta

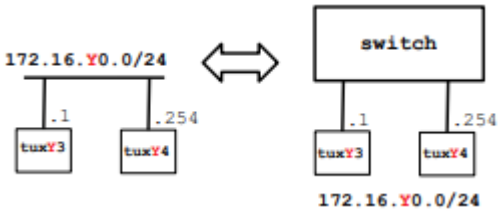
Esta struct é fundamental para iniciar a comunicação com o servidor FTP através da função connection_to_server() com o descritor de ficheiro socket_control. De seguida, a função read_response() verifica se há alguma resposta por parte do servidor com o socket_control (se houver,

buffer é esvaziado) e procede-se ao login() que ao receber a mensagem do servidor “Please specify the password”, valida .

De seguida é enviado o comando pasv para realizar um pedido ao servidor FTP, na função enter_passive_mode(), de forma a que o cliente seja responsável pela abertura da ligação TCP para os dados, ou seja, pela criação de uma nova ligação ao servidor através de um socket de dados . Posto isto, falta apena lançar o comando retr para inicializar a transferência de dados que ocorrem com as funções retrieve_file() e download(), cujo objetivo está acima descrito. Por fim, a função end_connection() fecha os sockets e liberta a memória alocada ao longo da execução, terminando o programa com sucesso.

Parte 2 – Configuração da rede

Experiência 1 – Configurar um IP de rede



O objetivo desta experiência é a configuração de dois computadores numa só rede. Para esse efeito, começamos por reiniciar o sistema através de **systemctl restart networking**. De seguida, ativámos as portas **eth0** através do comando **ifconfig eth0 up** e utilizámos o comando **ifconfig [ip]** para atribuir os endereços IP **172.16.20.1** e **172.16.20.254** aos computadores **tux23** e **tux24**, respetivamente. Por último, apagamos os endereços ARP com **arp -d [ip]**.

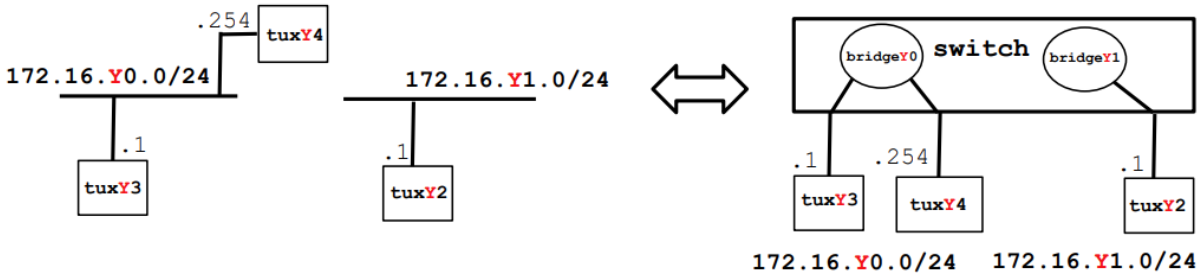
Para verificar a estabilidade desta ligação, servimo-nos do comando **ping 172.16.20.254** no **tux23**, que transmite pacotes ARP (*Address Resolution Protocol*), que verifica se um endereço IP está livre, e em caso afirmativo associa-o a um endereço MAC (identificador único do dispositivo físico numa rede local); assim, não têm endereço IP, já que são estes que o atribuem. Sendo este processo concluído com sucesso, os pacotes ICMP são transmitidos de um computador para outro.

Os pacotes ARP enviam os endereços IP e MAC e pedem o IP para o qual querem reencaminhar a informação. Após o computador com esse IP receber a trama, este responde com o seu endereço MAC para começar a troca de informação.

A distinção entre os diferentes tipos de pacotes é realizada pela verificação do *header* da trama Ethernet. Esta tem também um campo *length*, que informa o recetor do tamanho da trama.

A interface *loopback* é uma interface de rede virtual que o computador utiliza de modo a ter um endereço IP no *router*, evitando assim a necessidade de uma interface física para realizar testes de diagnóstico de maneira a verificar se a ligação se encontra ativa.

Experiência 2 – Implementar duas bridges no switch

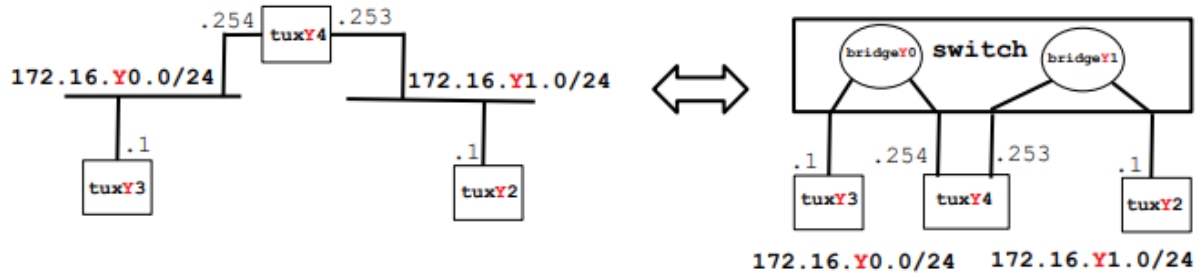


Esta experiência consiste em criar duas *bridges* no *switch*, uma com os computadores **tux23** e **tux24** e outra com o **tux22**, criando assim duas sub-redes diferentes. Assim, este último deixa de ter acesso aos outros dois dispositivos

Para esse efeito, abrimos a aplicação GTKTerm, colocamos a Baudrate em 115200bps e reiniciamos o *switch* com **/system reset-configuration** e executamos o comando **/interface bridge add name=bridge20** e **/interface bridge add name=bridge21**. De seguida, removemos os *ports* utilizados pelo **tux22**, **tux23** e **tux24** da *default bridge* e atribuímo-los às *bridges* correspondentes (ver imagem), com o comando **/interface bridge port remove** e **/interface bridge port add bridge=bridge20 interface=etherZ** (Z corresponde às portas onde os computadores estão ligados), respetivamente.

Em **tux23**, usar o comando **ping** para transmitir pacotes aos **tux24** e **tux22**, com endereços 172.16.20.254 e 172.16.20.1. Por último, usar **ping -b 172.16.20.255** em **tux23** e **tux22** para transmitir em *broadcast*. Analisando o resultado, podemos concluir que existem dois domínios de *Broadcast*, já que o *Broadcast* em **tux22** não é detetado em **tux23** e vice-versa.

Experiência 3 – Configurar um router em Linux



Com esta experiência, configurámos o **tux24** como um *router*, ligando a este as duas *bridges* criadas anteriormente. Assim, possibilitamos a comunicação entre ambas.

Para começar, configuramos a porta eth1 do tux24, usando a gama de endereços 172.16.21.253/24 e adicionamo-la à bridge 21, seguindo as instruções dos guiões anteriores e da imagem acima, acrescentando apenas os comandos **echo 1 > /proc/sys/net/ipv4/ip_forward** e **echo 0 > /proc/sys/net/ipv4/icmp_echo_ignore_broadcasts** para ativar IP Forwarding e desativar ICMP *echo ignore broadcast*, respetivamente.

Continuamos reconfigurando os dispositivos **tux23** e **tux22**, adicionando-lhes as rotas de cada um, com o comando **route add -net 172.16.20.0/24 gw 172.16.20.254** e **route add -net 172.16.21.0/24**

gw 172.16.20.253. O primeiro valor representa a gama de endereços atribuída à rota, e o segundo o endereço para o qual se reencaminham os dados.

Concluídas estas etapas, é possível finalmente aos computadores comunicarem entre si. Para testarmos esta ligação, executamos o comando **ping** no **tux23** para os vários PC's. De seguida, apagamos os registos nas tabelas ARP em todos os dispositivos, com **arp -d [ip]**, trocamos para o tux24 e utilizamos o **ping** para o **tux22**.

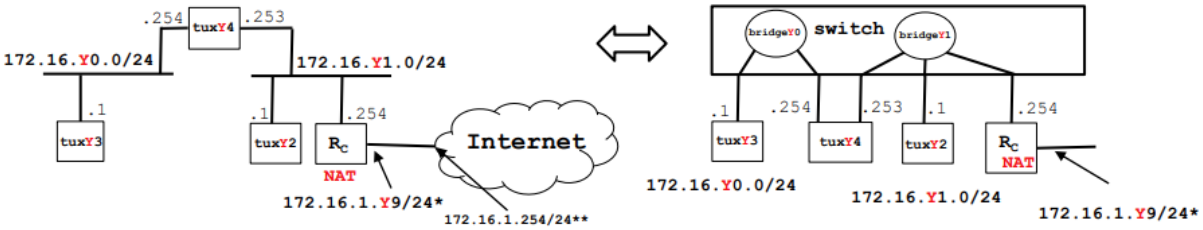
Podemos concluir que os pedidos do **tux23** par o **tux22** são transmitidos para o IP 172.16.20.254 do **tux24**, que os reencaminha para o **tux22** através da porta **eth1**. O inverso acontece nas comunicações do **tux22** para o **tux23**. Ao analisar os *logs*, percebemos que os pacotes ICMP têm como destino o endereço MAC do **tux24**, e, na resposta do outro computador, o pacote contém como origem o mesmo endereço MAC.

Os pacotes ARP enviam um *broadcast* para a rede local, inquirindo qual dos computadores tem o endereço que corresponde ao IP ao qual o pacote deve ser reencaminhado. Os endereços MAC apresentados são os de origem.

O ICMP (*Internal Control Message Protocol*) é usado como forma de diagnóstico de erros de comunicação em determinada rede. Tanto os pacotes *reply* como os *request* são verificados ao longo de toda a rede e de todos os dispositivos. Entre comunicações do **tux22** e **tux23**, estes, como todo o tráfego, passam primeiro pelo **tux24**.

Uma *forwarding table* contém um IP de destino, para onde o pacote em questão deverá ser mandado, , a *gateway*, ou seja, o próximo ponto por onde a informação passa, a *mask*, que é usada para saber quais dos *bits* correspondem à rede e sub-rede, e quais referem o *host*, bem como as *flags*, que dão informações sobre a rota. Por último, tem ainda a *metric*, que determina o custo da rota, a *ref*, que mostra o número de referências da rota, o *use*, que conta o número de falhas em. pesquisa durante a rota, e ainda a interface utilizada (por exemplo, **eth0**).

Experiência 4 – Configurar um router comercial e implementar o NAT



Esta experiência tem como objetivo a configuração de um *router* comercial com implementação NAT (*Network Address Translation*), de forma a possibilitar a comunicação com redes externas. Este protocolo reescreve os endereços IP de origem de uma rede interna de forma a comunicarem com uma externa. Sem este, os IP's da rede interna não são reconhecidos fora desta.

O NAT gera um número de 16 *bits*, utilizando uma *hash table*, e associa-o a uma porta de origem; aquando da resposta a determinado pacote, o processo é revertido. Desta forma, não é preciso registar um IP privado na Internet sem necessidade; sempre que este precise de comunicar com

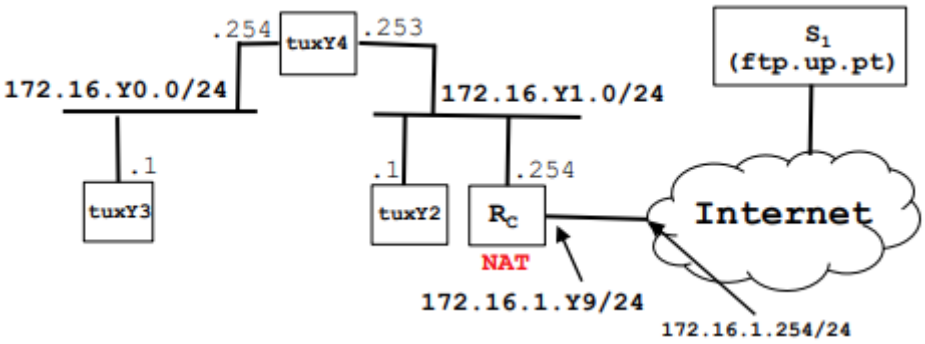
recursos externos, utiliza este protocolo para comunicar com outras redes. Em suma, permite que um único endereço IP represente uma rede privada na Internet, e assim comunicar com oenterior

Para o configurar, começamos por adicionar a porta do *router* no *switch* à *bridge* 21. No GTKTerm, executamos o comando **/ip address add address=172.16.1.29/24 interface=ether1** e /**ip address add address=172.16.21.254/24 interface=ether2**, respetivamente, a *address* do *router* para a Internet e para a rede pública; para de seguida, adicionamos as rotas para o **tux24**, com **route add default gw 172.16.20.254**, em **tux23**, e para o *router*, com **route add default gw 172.16.21.254**, em **tux22** e **tux24**. Por último, configuramos a gamas de endereços do *router*, executando **/ip route add dst-address=172.16.20.0/24 gateway=172.16.21.253** no GTKTerm. Para confirmar o que fizemos anteriormente, usámos o **ping** com os endereços do **tux22**, **tux24** e *router*.

Continuamos no **tux22**, escrevendo os comandos **echo 0 > /proc/sys/net/ipv4/conf/eth0/accept_redirects** e **echo 0 > /proc/sys/net/ipv4/conf/all/accept_redirects**, removendo também a rota para o **tux23** pelo **tux24** (**route del -net 172.16.20.0/24**) e tentamos, no computador onde estávamos, dar **ping** no primeiro. Como o caminho não se encontra disponível, os pacotes serão reencaminhados para a *default gateway*. Continuamos executando **traceroute** de **tux22** para **tux23**. Voltamos a adicionar o caminho previamente removido e voltamos a repetir o comando anterior. Para ativar os redireccionamentos, repetimos os dois comandos **echo** anteriores, mas com o número 1 em vez de 0, verificando que os pacotes são reencaminhados para a *default gateway* quando a rota pedida não é possível.

Concluindo a experiência, verificamos que todos os computadores conseguem transmitir pacotes ao *router*, sendo que os que têm como origem o **tux23** são reencaminhados primeiro para o **tux24**. Por último, desativamos o NAT (/ip firewall nat disable 0) e tentamos executar **ping** do **tux24** para o *router*, o que nos permite aferir que não é possível comunicar com o router sem *NAT*, pois este não consegue identificar o computador de origem.

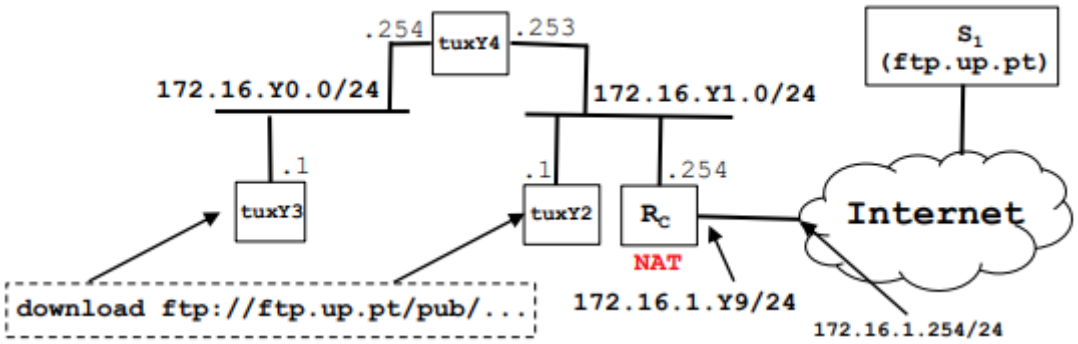
Experiência 5 – DNS



Esta experiência consiste em configurar o DNS (*Domain Name System*), para conseguirmos aceder à Internet através da rede por nós criada. Para o efeito, editámos o ficheiro resolv.conf, que se encontra em /etc escrevendo "nameserver 172.16.1.1". Também foi necessário reativar o protocolo NAT com **/ip firewall nat add chain=srcnat action=masquerade out-interface=ether1**.

Para testar a ligação realizámos um **ping** a www.google.pt; com este, conseguimos perceber que, ao fazer uma solicitação a um servidor externo, é enviado um pacote DNS que pede o endereço IP do mesmo, enviando este, como resposta, a informação indicada.

Experiência 6 – Ligações TCP



Para concluir o projeto, executámos a aplicação de download desenvolvida na primeira parte do projeto. O teste foi realizado através do servidor FTP da Universidade do Porto (ftp.up.pt), tendo sido bem-sucedida a transferência de um ficheiro que nos foi proposto, sem apresentar nenhum erro digno de nota.

Concluímos que o servidor FTP abriu duas conexões TCP, uma para enviar comandos e outra para a receção de dados. Para criar conexões TCP bastam apenas três passos, definir a porta, estabelecer a conexão ao servidor com essa porta, e claro quando não for mais necessária a conexão, encerrá-la.

O mecanismo ARQ (*Automatic Repeat Request*) utilizado pelo TCP, funciona com o método da janela deslizante, que usa *acknowledgment numbers*, indicadores da correta receção da trama, *window size* para a gama de pacotes que o emissor pode enviar e o *sequence number* que contém o número do pacote a ser enviado. Este mecanismo faz a gestão de erros aquando da transmissão de dados, e por isso quando falha o envio de uma trama, o recetor continua a receber e processar as tramas seguintes, sendo que têm sempre contido no ACK o número da trama que falhou primeiro. No final do envio, o emissor verifica os ACK e reenvia as tramas perdidas.

Durante a transferência, verificámos que o recetor envia, em cada pacote ACK, o tamanho da janela TCP, mostrando assim que recebeu os dados. Esta janela pode ficar impossibilitada de receber mais dados caso não esteja a ler os pacotes à velocidade necessária. Se tal acontecer, o recetor envia uma mensagem ACK com o bit *Window Full* ativo, o que faz com que pare de receber dados temporariamente. Desta forma, evitamos que o recetor rejeite pacotes e mantemos a taxa de retransmissão mínima. Quando for possível continuar o envio, o recetor envia um *TCP Window Update*. Podemos verificar o controlo de congestão nos *screenshots* dos *logs* da experiência 6 e no primeiro gráfico apresentado.

O *Sequence Number Field* identifica o primeiro byte da mensagem, enquanto o *ACK Number Field* revela o *Sequence Number* do byte que se espera receber a seguir.

Quando corremos o nosso programa em dois computadores diferentes ao mesmo tempo, abrindo assim duas conexões TCP, ocorre uma queda na taxa de transmissão, já que a taxa de transferência se mantém a mesma, mas é repartida de forma igual para as duas ligações. O gráfico relativo a esta situação é o segundo.

Conclusões

Através deste trabalho, foi-nos possível consolidar os conhecimentos apresentados ao longo da UC.

A aplicação de download permitiu-nos entender melhor o protocolo FTP (*File Transfer Protocol*) e conceitos associados.

A configuração de rede permitiu-nos entender as matérias abordadas, passo a passo, e entender melhor esta realidade com a qual temos contacto no quotidiano. Ambas as partes foram concluídas com sucesso.

Em suma, sentimo-nos realizados com a conclusão do projeto, já que cumprimos os requisitos propostos pelo professor, e permitiu-nos uma melhor compreensão dos temas abordados.

Anexos

Código da aplicação

parsing.h

```
#include <arpa/inet.h>
#include <stdlib.h>

#define FALSE 0
#define TRUE 1

typedef struct arguments{
    char user[256];
    char password[256];
    char host[256];
    char url_path[256];
    char file_name[256];
    char host_name[256];
    char ip[256];
} arguments;

/**
 * @brief Parses the URL from arguments to the args struct
 *
 * @param url string received through the arguments of download
 * @param arguments struct where we will save the information
 * @return int Returns 0 if there are no errors or a positive number otherwise
 */
int parse_args(char *url, arguments *arguments);
/**
 * @brief Given a host name returns the ip address
 *
 * @param host String with the host's name
```

```
 * @param arguments struct args in order to put the IP adress
 * @return int Returns 0 if there are no errors or 1 otherwise
 */
int get_ip(char *host, arguments *arguments);
/**
 * @brief Gets the last element of the path, which is the file name
 *
 * @param arguments arguments struct to get path and fill with the filename
 * @return int Returns 0 if there are no errors
 */
int get_filename(arguments *arguments);

#endif
```

parsing.c

```
#include "parsing.h"

int parse_args(char *url, arguments *arguments){
    char ftp[7]= "ftp://";
    int url_len = strlen(url);
    int i = 0;
    int j = 0;
    int user_flag = FALSE;
    int password_flag = FALSE;
    int host_flag = FALSE;
    int path_flag = FALSE;

    while(i < url_len){
        if(url[i] == ftp[i] && i>=0 && i<= 5){
            i++;
            user_flag = TRUE;
        }
        else if(i > 5 && url[i] != ':' && url[i] != '@' && url[i] != '/'){
            if(user_flag){
              arguments->user[j] = url[i];
              i++;
              j++;
            } else if(password_flag){
                arguments->password[j] = url[i];
                j++;
                i++;
            } else if(host_flag){
                arguments->host[j] = url[i];
                i++;
                j++;
            } else if (path_flag) {
                arguments->url_path[j] = url[i];
```

```
            i++;
            j++;
        }

    }
    else if(url[i] == ':'){
        password_flag = TRUE;
        user_flag = FALSE;
        j = 0;
        i++;
    }
    else if(url[i] == '@'){
        password_flag = FALSE;
        host_flag = TRUE;
        j=0;
        i++;
    }
    else if(url[i] == '/'){
        if (user_flag) {
            //it means that the user was not specified
            for (int k = 0; k < j; k++) {
                arguments->host[k] = arguments->user[k];
            }
            host_flag = FALSE;
            user_flag = FALSE;
            path_flag = TRUE;
            j=0;
            i++;
            char def_user[] = "anonymous";
            char def_password[] = "pass";
            for (int k = 0; k < 10; k++) {
                arguments->user[k] = def_user[k];
            }
            for (int k = 0; k < 5; k++) {
                arguments->password[k] = def_password[k];
            }
        } else if (host_flag) {
            host_flag = FALSE;
            j = 0;
            i++;
            path_flag = TRUE;
        } else
        {
            arguments->url_path[j] = url[i];
            i++;
            j++;
        }
```

```
        }
        else {
            break;
        }
    }
    if(get_ip(arguments->host, arguments) != 0){
        printf("getIp() failed\n");
        return -1;
    }

    if(get_filename(arguments) != 0){
        printf("Error getFileName()\n");
        return -1;
    }

    return 0;

}

int get_ip(char *host, arguments *arguments){
    struct hostent *h;

    if ((h = gethostbyname(host)) == NULL) {
        herror("gethostbyname()");
        exit(-1);
    }

    strcpy(arguments->host_name,h->h_name);
    strcpy(arguments->ip,inet_ntoa(*((struct in_addr *) h->h_addr)));

    return 0;
}

int get_filename(arguments *arguments){

    char path_cpy[256];

    strcpy(path_cpy,arguments->url_path);

    int k = 0;
    for(int i = strlen(path_cpy)-1; i>=0; i--){
        if(path_cpy[i] == '/'){
            for(int j = i+1; j < strlen(path_cpy)-1; j++){
                strcpy(&arguments->file_name[k],&path_cpy[j]);
                k++;
            }
            break;
        }
```

```
    }

    return 0;
}
```

TCP.h

```
#ifndef __TCP_H
#define __TCP_H

#include <stdio.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <strings.h>
#include <arpa/inet.h>
#include <string.h>

typedef struct TCP {
    int socket_control;
    int socket_data;
    char new_ip[1024];
    int new_port;
} TCP;

/**
 * @brief creates control socket (setting socket_fd) and connects to ftp
server
 *
 * @param ip Server IP Adress
 * @param port Server Port
 * @return int Returns 0 if there are no errors or a positive number otherwise
 */
int connection_to_server(char * ip, int port);

/**
 * @brief writes cmd to the server through socket_fd
 *
 * @param socket_fd File Descriptor of the socket
 * @param comd Command to send
 * @return int Returns 0 if there are no errors or a positive number otherwise
 */
int write_to_server(int socket_fd, char * cmd);
```

```
/**
 * @brief Reads reply from server
 *
 * @param socket_fd File Descriptor of the socket
 * @param buf Buffer writen by the server with the response that is going to
be read
 * @return int Returns 0 if there are no errors, 1 otherwise (return value is
boolean)
 */
int read_response(int socket_fd, char * buf);

/**
 * @brief Enables the user's login
 *
 * @param user String containing the user's name
 * @param password String containing the user's password
 * @param socket_fd File Descriptor of the socket
 * @return int Returns 0 if there are no errors, -1 otherwise
 */
int login(char * user, char * password,int socket_fd);

/**
 * @brief Creates the new ip adress and port
 *
 * @param socket_fd File Descriptor of the socket
 * @param ip Server IP Adress
 * @return int Returns the new port created, -1 otherwise
 */
int enter_passive_mode(int socket_fd,char *ip);

/**
 * @brief Receives the file and after reading it, it returns its size
 *
 * @param socket_fd File Descriptor of the socket
 * @param command Command to be wrtiten in socket_fd
 * @return int Returns the size of the file, -1 if encounters errors
 */
int retrieve_file(int socket_fd, char *command);

/**
 * @brief Enables the download of the file
 *
 * @param filename Name of the file
 * @param socketdata File Descriptor of the socket
 * @param socket_fd File Descriptor of the socket
 * @param file_size Size of the file
 * @return int Returns 0 if there are no errors, -1 otherwise
 */
```

```
int download(char * filename, int socketdata, int socket_fd, int file_size);

/**
 * @brief Ends connection with the server
 *
 * @param socket_fd File Descriptor of the socket
 * @param socketdata File Descriptor of the socket
 * @return int Returns 0 if there are no errors, -1 otherwise
 */
int end_connection(int socket_fd,int socketdata);

#endif
```

TCP.c

```
#include "TCP.h"

int connection_to_server(char * ip, int port){
    int sockfd;
    struct sockaddr_in server_addr;

    /*server address handling*/
    bzero((char *) &server_addr, sizeof(server_addr));
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = inet_addr(ip);    /*32 bit Internet address
network byte ordered*/
    server_addr.sin_port = htons(port);        /*server TCP port must be
network byte ordered */

    /*open a TCP socket*/
    if ((sockfd = socket(AF_INET, SOCK_STREAM, 0)) < 0) {
        perror("socket()");
        exit(-1);
    }
    /*connect to the server*/
    if (connect(sockfd,(struct sockaddr *) &server_addr,sizeof(server_addr)) <
0) {
        perror("connect()");
        return -1;
    }
    return sockfd;
}

int write_to_server(int sock_fd, char * cmd){
    int bytes_written= 0;
    bytes_written = write(sock_fd, cmd, strlen(cmd));
    if (!(bytes_written > 0)){
        perror("write() command not sent");
```

```
        return -1;
    }
    return bytes_written;
}

int read_response(int socket_fd, char * buf){
    FILE * socket = fdopen(socket_fd, "r");

    size_t bytes_read = 0;

    do {
        getline(&buf, &bytes_read,socket);
        printf("%s", buf);
    } while ( !('1' <= buf[0] && buf[0] <= '5') || buf[3] != ' ');
    if (buf[3] == ' '){
      int code=0;
      if (code == 550 || code == 530)
      {
        printf("Command error\n");
        return 1;
      }
    }
    return 0;
}

int login(char * user, char * password, int socket_fd){
    char buf[1024];

    char *user_cmd = malloc(5 + strlen(user) + 3);
    user_cmd[0] = '\0';
    strcat(user_cmd, "user ");
    strcat(user_cmd, user);
    strcat(user_cmd, "\r\n");

    char *password_cmd = malloc(5 + strlen(password) + 3);
    password_cmd[0] = '\0';
    strcat(password_cmd, "pass ");
    strcat(password_cmd, password);
    strcat(password_cmd, "\r\n");


    if (write_to_server(socket_fd, user_cmd) < 0) {
        printf("Error writing USER command\n");
        return -1;
    }
    if (read_response(socket_fd, buf)) {
        printf("Error reading reply to USER command\n");
        return -1;
    }
```

```
    if (write_to_server(socket_fd, password_cmd) < 0) {
        printf("Error writing PASS command\n");
        return -1;
    }

    if (read_response(socket_fd, buf)) {
        printf("Error reading reply to PASS command\n");
        return -1;
    }
    return 0;
}
int enter_passive_mode(int socket_fd, char *ip){

    char buff[1024];
    if (write_to_server(socket_fd, "PASV\r\n") < 0) {
        printf("Error writing PASV\n");
        return -1;
    }

    read(socket_fd, buff, 1024);

    //parse resposta
    strtok(buff, "(");

    char* ip1 = strtok(NULL, ",");
    char* ip2 = strtok(NULL, ",");
    char* ip3 = strtok(NULL, ",");
    char* ip4 = strtok(NULL, ",");

    char* new_ip = malloc(256);
    new_ip[0] = '\0';
    strcat(new_ip, ip1);
    strcat(new_ip, ".");
    strcat(new_ip, ip2);
    strcat(new_ip, ".");
    strcat(new_ip, ip3);
    strcat(new_ip, ".");
    strcat(new_ip, ip4);

    char* p1 = strtok(NULL, ",");
    char* p2 = strtok(NULL, ")");

    for (size_t i = 0; i < strlen(new_ip); i++)
    {
        ip[i] = new_ip[i];
    }
```

```
    int new_port = atoi(p1)*256 + atoi(p2);

    return new_port;

    return -1;
}

int retrieve_file(int socket_fd, char *command){
    int size;
    char buffer[1024]= {0};

    if((size = write(socket_fd, command, strlen(command))) <= 0){
        printf("Error: command not sent\n");
        return -1;
    }

    read(socket_fd, buffer, 1024);

    printf("%s\n", buffer);

    strtok(buffer, "(");

    char* size_string = strtok(NULL, "bytes");

    return atoi(size_string);

    return -1;

}

int download(char * filename, int socketdata, int socket_fd, int file_size){
    int file_fd;

    if ((file_fd = open(filename, O_WRONLY | O_CREAT, 0777)) < 0) {
        fprintf(stderr, "Error opening data file!\n");
        return -1;
    }

    char buff[10024] = {0};
    int numBytesRead=0;

    while((numBytesRead = read(socketdata, buff, 10024)) > 0) {
        file_size -= numBytesRead;
        printf("saving... %d bytes left\n", file_size);
        if (write(file_fd, buff, numBytesRead) < 0) {
            fprintf(stderr, "Error writing data to file!\n");
            return -1;
        }
```

```
    }
    if(read_response(socket_fd, buff)) {
        printf("Error reading reply\n");
        return -1;
    }
    if (close(file_fd) < 0) {
        fprintf(stderr, "Error closing file!\n");
        return -1;
    }

    return 0;
}

int end_connection(int socket_fd, int socketdata){
    char buffer[1024];
    char quit_cmd[] = "quit\r\n";

    if(write_to_server(socket_fd, quit_cmd) < 0) {
        printf("Error writing QUIT command to server\n");
        return -1;
    }

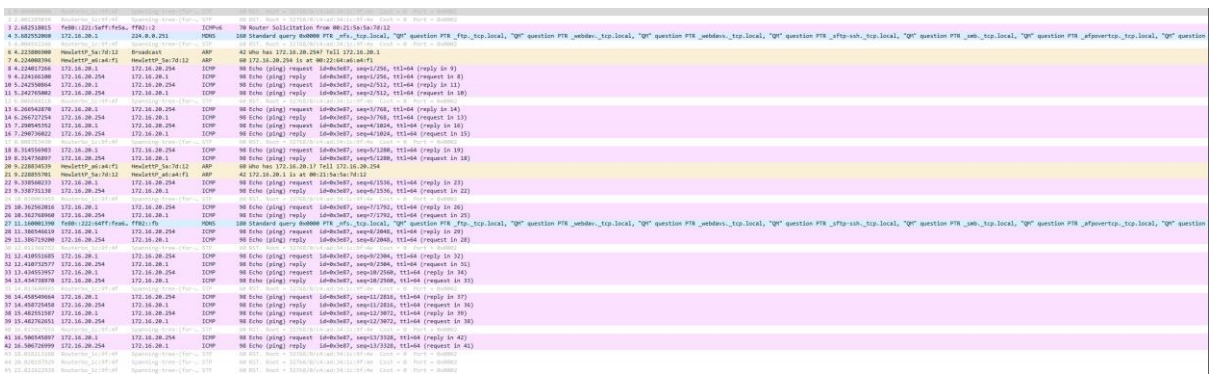
    if(read_response(socket_fd, buffer)) {
        printf("Error reading reply to QUIT command\n");
        return -1;
    }

    if(close(socket_fd) < 0 || close(socketdata) < 0){
        printf("Error closing sockets\n");
        return -1;
    }

    return 0;
}
```

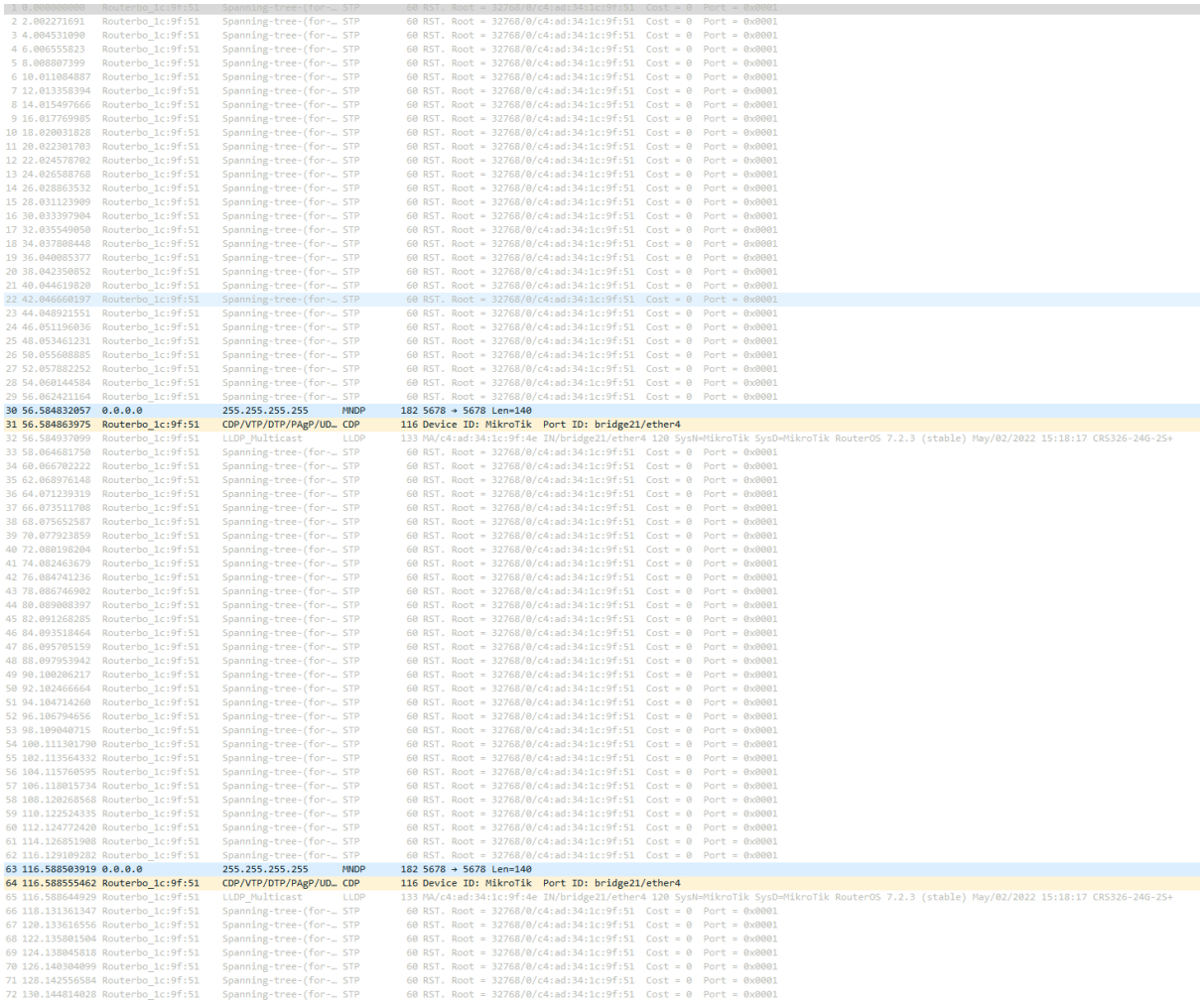
Logs

Experiência 1



Experiência 2

1	0.000000000	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
2	2.002186905	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
3	2.418307102	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x41ef, seq=1/256, ttl=64 (reply in 4)
4	2.418516560	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x41ef, seq=1/256, ttl=64 (request in 3)
5	3.446436372	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x41ef, seq=2/512, ttl=64 (reply in 6)
6	3.446623760	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x41ef, seq=2/512, ttl=64 (request in 5)
7	4.004460276	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
8	4.470454289	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x41ef, seq=3/768, ttl=64 (reply in 9)
9	4.470637277	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x41ef, seq=3/768, ttl=64 (request in 8)
10	6.006723519	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
11	7.510405388	HewlettP_5a:7d:12	HewlettP_a6:a4:f1	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
12	7.510598503	HewlettP_a6:a4:f1	HewlettP_5a:7d:12	ARP	60 172.16.20.254 is at 00:22:64:a6:a4:f1
13	7.653637361	HewlettP_a6:a4:f1	HewlettP_5a:7d:12	ARP	60 Who has 172.16.20.1? Tell 172.16.20.254
14	7.653658314	HewlettP_5a:7d:12	HewlettP_a6:a4:f1	ARP	42 172.16.20.1 is at 00:21:5a:5a:7d:12
15	8.008991302	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
16	10.011262158	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
17	12.013299670	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002

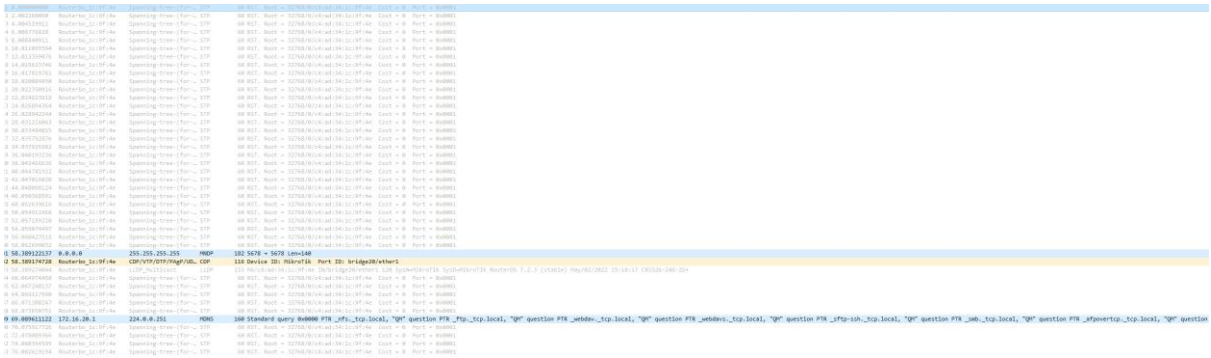


1	0.000000000	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
2	2.002264361	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
3	4.004297752	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
4	6.006568469	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
5	8.008842258	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
6	10.011109133	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
7	12.013250571	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
8	12.606268958	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
9	12.606294241	Routerbo_1c:9f:4f	CDP/VTP/DTP/PAgP/UD…	CDP	116	Device ID: MikroTik Port ID: bridge20/ether2
10	12.606368135	Routerbo_1c:9f:4f	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:9f:4e IN/bridge20/ether2 120 SysN=MikroTik SysD=MikroTik
11	14.015524919	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
12	16.017795146	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
13	18.020064745	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
14	20.022333506	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
15	22.024343500	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
16	24.026606464	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
17	26.028875015	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
18	28.031145661	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
19	30.033287658	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
20	32.035558095	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
21	34.037824481	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
22	36.040100924	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
23	38.042362002	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
24	40.044381285	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
25	42.046642363	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
26	44.048911264	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
27	46.051182189	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
28	48.053322300	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
29	50.055590223	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
30	52.057859472	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
31	54.060125300	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
32	56.062397483	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
33	58.064426893	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
34	60.066694606	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
35	62.068967767	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
36	64.071224654	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
37	66.073381038	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
38	68.075643723	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
39	70.077911785	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
40	72.080179149	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
41	72.602389105	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
42	72.602415296	Routerbo_1c:9f:4f	CDP/VTP/DTP/PAgP/UD…	CDP	116	Device ID: MikroTik Port ID: bridge20/ether2
43	72.602490167	Routerbo_1c:9f:4f	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:9f:4e IN/bridge20/ether2 120 SysN=MikroTik SysD=MikroTik
44	74.082451821	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
45	75.299730438	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=1/256, ttl=64 (no response found!)
46	75.299991300	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=1/256, ttl=64
47	76.084466354	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
48	76.325364932	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=2/512, ttl=64 (no response found!)
49	76.325586891	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=2/512, ttl=64
50	77.349374049	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=3/768, ttl=64 (no response found!)
51	77.349604110	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=3/768, ttl=64
52	78.086732810	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
53	78.373371991	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=4/1024, ttl=64 (no response found!)
54	78.373612598	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=4/1024, ttl=64
55	79.397367558	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=5/1280, ttl=64 (no response found!)
56	79.397611518	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=5/1280, ttl=64
57	80.089001431	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
58	80.340286578	HewlettP_a6:a4:f1	HewlettP_5a:7d:12	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
59	80.340304667	HewlettP_5a:7d:12	HewlettP_a6:a4:f1	ARP	42	172.16.20.1 is at 00:21:5a:5a:7d:12
60	80.421365011	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=6/1536, ttl=64 (no response found!)
61	80.421593257	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=6/1536, ttl=64
62	81.445373639	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=7/1792, ttl=64 (no response found!)
63	81.445636108	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=7/1792, ttl=64
64	82.091268516	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
65	84.093410931	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
66	86.095681368	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002

1	0.000000000	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
2	2.002273819	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
3	4.004296835	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
4	6.006556406	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
5	8.008837698	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
6	10.011106349	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
7	12.013260878	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
8	14.015529110	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
9	16.017804326	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
10	18.020083941	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
11	20.022350776	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
12	22.024388180	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
13	24.026666399	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
14	26.028925412	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
15	28.031210615	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
16	30.033356483	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
17	32.035626461	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
18	34.037904680	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
19	36.040171794	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
20	36.562133844	0.0.0.0	255.255.255.255	MNDP	182	5678 → 5678 Len=140
21	36.562176168	Routerbo_1c:9f:4e	CDP/VTP/DTP/PAgP/UD…	CDP	116	Device ID: MikroTik Port ID: bridge20/ether1
22	36.562261795	Routerbo_1c:9f:4e	LLDP_Multicast	LLDP	133	MA/c4:ad:34:1c:9f:4e IN/bridge20/ether1 120 SysN=MikroTik SysD=MikroTik
23	38.042452388	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
24	39.259905114	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=1/256, ttl=64 (no response found!)
25	39.259936333	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=1/256, ttl=64
26	40.044463671	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
27	40.285524577	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=2/512, ttl=64 (no response found!)
28	40.285534844	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=2/512, ttl=64
29	41.309547170	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=3/768, ttl=64 (no response found!)
30	41.309556110	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=3/768, ttl=64
31	42.046736303	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
32	42.333539173	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=4/1024, ttl=64 (no response found!)
33	42.333566761	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=4/1024, ttl=64
34	43.357549684	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=5/1280, ttl=64 (no response found!)
35	43.357565817	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=5/1280, ttl=64
36	44.049017525	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
37	44.300233402	HewlettP_a6:a4:f1	HewlettP_5a:7d:12	ARP	42	Who has 172.16.20.1? Tell 172.16.20.254
38	44.300405563	HewlettP_5a:7d:12	HewlettP_a6:a4:f1	ARP	60	172.16.20.1 is at 00:21:5a:5a:7d:12
39	44.381533724	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=6/1536, ttl=64 (no response found!)
40	44.381555654	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=6/1536, ttl=64
41	45.405566864	172.16.20.1	172.16.20.255	ICMP	98	Echo (ping) request id=0x4249, seq=7/1792, ttl=64 (no response found!)
42	45.405597315	172.16.20.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4249, seq=7/1792, ttl=64
43	46.051288550	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
44	48.053439447	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
45	50.055713825	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
46	52.057987225	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
47	54.060262022	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
48	56.062535562	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
49	58.064555295	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
50	60.066813330	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
51	62.069068501	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
52	64.071330727	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
53	66.073513681	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
54	68.075772903	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
55	70.078031008	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
56	72.080287087	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
57	74.082550639	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
58	76.084620240	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001
59	78.086878554	Routerbo_1c:9f:4e	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0001

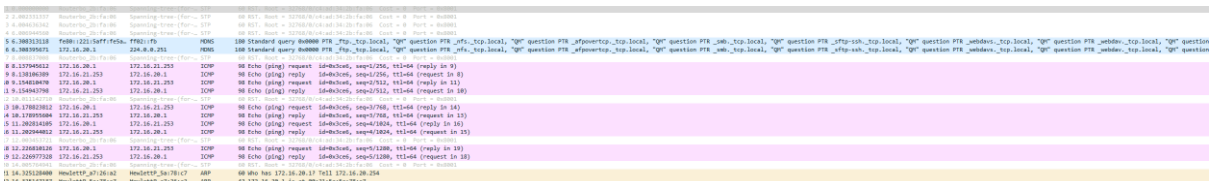
1	0.000000000	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
2	2.002250739	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
3	4.004508532	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
4	6.006755220	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
5	8.008843718	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
6	10.011109193	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
7	12.013372014	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
8	14.015632671	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
9	16.017799670	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
10	16.402907509	0.0.0.0	255.255.255.255	MNDP	182 5678 → 5678 Len=140
11	16.402951020	Routerbo_1c:9f:51	CDP/VTP/DTP/PAgP/UD…	CDP	116 Device ID: MikroTik Port ID: bridge21/ether4
12	16.403038461	Routerbo_1c:9f:51	LLDP_Multicast	LLDP	133 MA/c4:ad:34:1c:9f:4e IN/bridge21/ether4 120 SysN=MikroTik SysD=MikroTik Rc
13	18.020054110	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
14	20.022315256	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
15	22.024570953	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
16	24.026824555	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
17	26.028885606	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
18	28.031135227	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
19	30.033394906	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
20	32.035658495	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
21	34.037848333	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
22	36.040111922	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
23	37.466212858	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=1/256, ttl=64 (no response found!)
24	38.042370553	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
25	38.478853040	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=2/512, ttl=64 (no response found!)
26	39.502850856	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=3/768, ttl=64 (no response found!)
27	40.044639172	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
28	40.526848253	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=4/1024, ttl=64 (no response found!)
29	41.550843485	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=5/1280, ttl=64 (no response found!)
30	42.046903529	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
31	42.574852336	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=6/1536, ttl=64 (no response found!)
32	43.598849244	172.16.21.1	172.16.21.255	ICMP	98 Echo (ping) request id=0x2730, seq=7/1792, ttl=64 (no response found!)
33	44.048951031	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001
34	46.051227750	Routerbo_1c:9f:51	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:51 Cost = 0 Port = 0x0001

1	0.000000000	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
2	2.002271484	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
3	4.004529210	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
4	6.006792104	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
5	8.008953726	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
6	8.393802537	0.0.0.0	255.255.255.255	MNDP	182 5678 → 5678 Len=140
7	8.393847865	Routerbo_1c:9f:4f	CDP/VTP/DTP/PAgP/UD…	CDP	116 Device ID: MikroTik Port ID: bridge20/ether2
8	8.393937612	Routerbo_1c:9f:4f	LLDP_Multicast	LLDP	133 MA/c4:ad:34:1c:9f:4e IN/bridge20/ether2 120 SysN=MikroTik SysD=Mikr
9	10.011208937	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
10	12.013472041	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
11	14.015720896	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
12	16.017977574	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
13	18.020036318	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
14	20.022283428	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
15	22.024545484	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
16	24.026806981	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
17	26.028996121	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
18	28.031256780	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
19	30.033517299	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
20	32.035790460	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
21	34.038050490	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
22	36.040095545	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
23	38.042364236	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
24	40.044627619	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
25	42.046889954	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
26	44.049054510	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
27	46.051323550	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
28	48.053592520	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
29	50.055858627	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
30	52.058130181	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
31	54.059202748	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
32	56.061471998	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
33	58.063738454	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
34	60.066007494	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
35	62.068278071	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
36	64.070156411	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
37	66.071504647	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002
38	68.073765515	Routerbo_1c:9f:4f	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:1c:9f:4e Cost = 0 Port = 0x0002

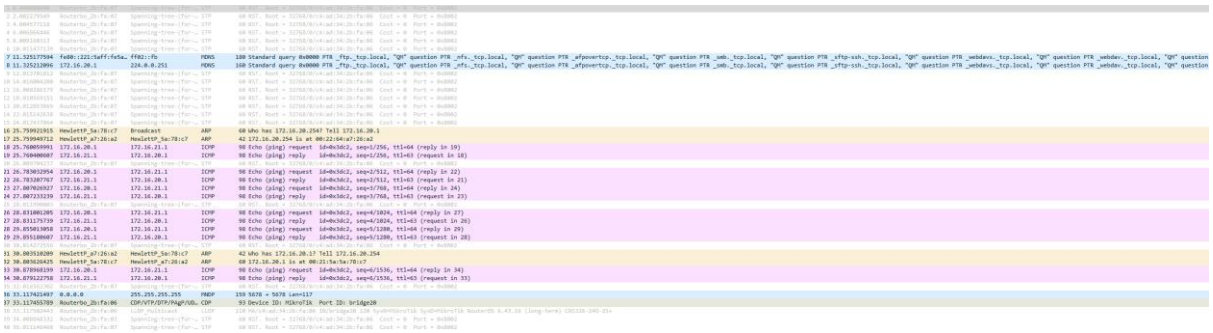


Experiência 3

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	1.741023767	HewlettP_5a:78:c7	Broadcast	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
3	1.741209338	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2
4	1.741227148	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x3c5c, seq=1/256, ttl=64 (reply in 5)
5	1.741348883	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3c5c, seq=1/256, ttl=64 (request in 4)
6	2.002307120	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
7	2.759148937	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x3c5c, seq=2/512, ttl=64 (reply in 8)
8	2.759281567	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3c5c, seq=2/512, ttl=64 (request in 7)
9	3.783140568	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x3c5c, seq=3/768, ttl=64 (reply in 10)
10	3.783306792	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3c5c, seq=3/768, ttl=64 (request in 9)
11	4.004633516	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
12	4.807134573	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x3c5c, seq=4/1024, ttl=64 (reply in 13)
13	4.807267553	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3c5c, seq=4/1024, ttl=64 (request in 12)
14	5.831116146	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x3c5c, seq=5/1280, ttl=64 (reply in 15)
15	5.831240186	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3c5c, seq=5/1280, ttl=64 (request in 14)
16	6.006937422	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
17	6.776307944	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 Who has 172.16.20.1? Tell 172.16.20.254
18	6.776320934	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 172.16.20.1 is at 00:21:5a:5a:78:c7
19	7.999234134	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
20	10.001550611	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
21	12.003853748	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
22	12.193121735	0.0.0.0	255.255.255.255	MNDP	159 5678 → 5678 Len=117
23	12.193158891	Routerbo_2b:fa:06	CDP/VTP/DTP/PAgP/UD…	CDP	93 Device ID: MikroTik Port ID: bridge20
24	12.193207151	Routerbo_2b:fa:06	LLDP_Multicast	LLDP	110 MA/c4:ad:34:2b:fa:06 IN/bridge20 120 SysN=MikroTik SysD=MikroTik Rou
25	14.006188732	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
26	16.008501507	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
27	18.010807156	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001



1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002307445	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	3.994627530	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
4	5.996935812	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
5	6.138801855	0.0.0.0	255.255.255.255	MNDP	159 5678 → 5678 Len=117
6	6.138837404	Routerbo_2b:fa:06	CDP/VTP/DTP/PAgP/UD…	CDP	93 Device ID: MikroTik Port ID: bridge20
7	6.138884757	Routerbo_2b:fa:06	LLDP_Multicast	LLDP	110 MA/c4:ad:34:2b:fa:06 IN/bridge20 120 SysN=MikroTik SysD=MikroTik Rou
8	6.447465844	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3d12, seq=1/256, ttl=64 (reply in 9)
9	6.447758482	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3d12, seq=1/256, ttl=63 (request in 8)
10	7.455969125	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3d12, seq=2/512, ttl=64 (reply in 11)
11	7.456231732	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3d12, seq=2/512, ttl=63 (request in 10)
12	7.999256316	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
13	8.479944400	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3d12, seq=3/768, ttl=64 (reply in 14)
14	8.480232429	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3d12, seq=3/768, ttl=63 (request in 13)
15	9.503958647	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3d12, seq=4/1024, ttl=64 (reply in 16)
16	9.504225165	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3d12, seq=4/1024, ttl=63 (request in 15)
17	10.001556705	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
18	10.527954944	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3d12, seq=5/1280, ttl=64 (reply in 19)
19	10.528242205	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3d12, seq=5/1280, ttl=63 (request in 18)
20	11.583930638	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
21	11.584051325	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2
22	11.993870363	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001



1	0.000000000	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
2	2.002254477	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
3	4.004524807	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
4	6.006820351	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
5	8.009112681	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
6	10.011397330	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
7	12.013678067	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
8	14.015962854	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
9	16.018260842	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
10	17.941169405	EncoreNe_c8:7c:55	Broadcast	ARP	42 Who has 172.16.21.1? Tell 172.16.21.253
11	17.941329272	HewlettP_5a:76:a8	EncoreNe_c8:7c:55	ARP	60 172.16.21.1 is at 00:21:5a:5a:76:a8
12	17.941347011	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=1/256, ttl=63 (reply in 13)
13	17.941482573	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=1/256, ttl=64 (request in 12)
14	18.020544582	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
15	18.964149841	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=2/512, ttl=63 (reply in 16)
16	18.964282330	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=2/512, ttl=64 (request in 15)
17	19.988143814	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=3/768, ttl=63 (reply in 18)
18	19.988307801	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=3/768, ttl=64 (request in 17)
19	20.022829300	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
20	21.012117952	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=4/1024, ttl=63 (reply in 21)
21	21.012250092	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=4/1024, ttl=64 (request in 20)
22	21.948689914	fe80::221:5aff:fe5a…	ff02::2	ICMPv6	70 Router Solicitation from 00:21:5a:5a:76:a8
23	22.025111783	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
24	22.036130503	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=5/1280, ttl=63 (reply in 25)
25	22.036259849	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=5/1280, ttl=64 (request in 24)
26	22.972642751	HewlettP_5a:76:a8	EncoreNe_c8:7c:55	ARP	60 Who has 172.16.21.253? Tell 172.16.21.1
27	22.972661957	EncoreNe_c8:7c:55	HewlettP_5a:76:a8	ARP	42 172.16.21.253 is at 00:e0:7d:c8:7c:55
28	23.060080197	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x3dc2, seq=6/1536, ttl=63 (reply in 29)
29	23.060209403	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x3dc2, seq=6/1536, ttl=64 (request in 28)
30	24.027396432	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
31	25.298700206	0.0.0.0	255.255.255.255	MNDP	159 5678 → 5678 Len=117
32	25.298731844	Routerbo_2b:fa:05	CDP/VTP/DTP/PAgP/UD…	CDP	93 Device ID: MikroTik Port ID: bridge21
33	25.298780104	Routerbo_2b:fa:05	LLDP_Multicast	LLDP	110 MA/c4:ad:34:2b:fa:05 IN/bridge21 120 SysN=MikroTik SysD=MikroTik Ro
34	26.029697492	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
35	28.031993455	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002
36	30.034280547	Routerbo_2b:fa:04	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:05 Cost = 0 Port = 0x8002

Experiência 4

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	1.774354789	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x45c1, seq=1/256, ttl=64 (reply in 3)
3	1.774681301	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45c1, seq=1/256, ttl=63 (request in 2)
4	2.002074503	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
5	2.787684050	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x45c1, seq=2/512, ttl=64 (reply in 6)
6	2.787983254	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45c1, seq=2/512, ttl=63 (request in 5)
7	3.811684703	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x45c1, seq=3/768, ttl=64 (reply in 8)
8	3.811957647	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45c1, seq=3/768, ttl=63 (request in 7)
9	4.004113596	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
10	4.835677744	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x45c1, seq=4/1024, ttl=64 (reply in 11)
11	4.835977576	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45c1, seq=4/1024, ttl=63 (request in 10)
12	5.859678956	172.16.20.1	172.16.21.1	ICMP	98 Echo (ping) request id=0x45c1, seq=5/1280, ttl=64 (reply in 13)
13	5.859947010	172.16.21.1	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45c1, seq=5/1280, ttl=63 (request in 12)
14	6.006225185	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
15	6.738145811	fe80::222:64ff:fea7…	ff02::2	ICMPv6	70 Router Solicitation from 00:22:64:a7:26:a2

1	0.000000000	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x45db, seq=1/256, ttl=64 (reply in 2)
2	0.000163640	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45db, seq=1/256, ttl=64 (request in 1)
3	0.272489512	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
4	1.021040044	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x45db, seq=2/512, ttl=64 (reply in 5)
5	1.021202008	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45db, seq=2/512, ttl=64 (request in 4)
6	2.045017021	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x45db, seq=3/768, ttl=64 (reply in 7)
7	2.045153492	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45db, seq=3/768, ttl=64 (request in 6)
8	2.274543690	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
9	3.069022283	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x45db, seq=4/1024, ttl=64 (reply in 10)
10	3.069159244	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45db, seq=4/1024, ttl=64 (request in 9)
11	4.093030898	172.16.20.1	172.16.20.254	ICMP	98 Echo (ping) request id=0x45db, seq=5/1280, ttl=64 (reply in 12)
12	4.093166183	172.16.20.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45db, seq=5/1280, ttl=64 (request in 11)
13	4.276629857	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
14	5.195803793	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 Who has 172.16.20.1? Tell 172.16.20.254
15	5.195823488	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 172.16.20.1 is at 00:21:5a:5a:78:c7
16	5.212988175	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
17	5.213099783	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2
18	6.278765960	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002108167	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	2.122831377	172.16.20.1	172.16.21.254	ICMP	98 Echo (ping) request id=0x45f7, seq=1/256, ttl=64 (reply in 4)
4	2.123178562	172.16.21.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45f7, seq=1/256, ttl=63 (request in 3)
5	3.151708898	172.16.20.1	172.16.21.254	ICMP	98 Echo (ping) request id=0x45f7, seq=2/512, ttl=64 (reply in 6)
6	3.152008800	172.16.21.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45f7, seq=2/512, ttl=63 (request in 5)
7	4.004177571	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
8	4.175710319	172.16.20.1	172.16.21.254	ICMP	98 Echo (ping) request id=0x45f7, seq=3/768, ttl=64 (reply in 9)
9	4.175974462	172.16.21.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45f7, seq=3/768, ttl=63 (request in 8)
10	5.199700426	172.16.20.1	172.16.21.254	ICMP	98 Echo (ping) request id=0x45f7, seq=4/1024, ttl=64 (reply in 11)
11	5.199970576	172.16.21.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45f7, seq=4/1024, ttl=63 (request in 10)
12	6.006255705	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
13	6.223696539	172.16.20.1	172.16.21.254	ICMP	98 Echo (ping) request id=0x45f7, seq=5/1280, ttl=64 (reply in 14)
14	6.223966479	172.16.21.254	172.16.20.1	ICMP	98 Echo (ping) reply id=0x45f7, seq=5/1280, ttl=63 (request in 13)
15	8.008328672	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
16	8.271666137	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
17	8.271802539	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2

1	0.000000000	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
2	1.371113326	172.16.21.1	172.16.20.1	UDP	74 56275 → 33434 Len=32
3	1.371138050	172.16.21.1	172.16.20.1	UDP	74 39864 → 33435 Len=32
4	1.371156069	172.16.21.1	172.16.20.1	UDP	74 60554 → 33436 Len=32
5	1.371173320	172.16.21.1	172.16.20.1	UDP	74 48064 → 33437 Len=32
6	1.371191270	172.16.21.1	172.16.20.1	UDP	74 49637 → 33438 Len=32
7	1.371208451	172.16.21.1	172.16.20.1	UDP	74 56286 → 33439 Len=32
8	1.371303715	172.16.21.254	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
9	1.371315449	172.16.21.1	172.16.20.1	UDP	74 45633 → 33440 Len=32
10	1.371317195	172.16.21.1	172.16.20.1	UDP	74 51545 → 33441 Len=32
11	1.371318243	172.16.21.1	172.16.20.1	UDP	74 60458 → 33442 Len=32
12	1.371319360	172.16.21.1	172.16.20.1	UDP	74 51772 → 33443 Len=32
13	1.371320338	172.16.21.1	172.16.20.1	UDP	74 54028 → 33444 Len=32
14	1.371321385	172.16.21.1	172.16.20.1	UDP	74 37192 → 33445 Len=32
15	1.371322363	172.16.21.1	172.16.20.1	UDP	74 46099 → 33446 Len=32
16	1.371339684	172.16.21.1	172.16.20.1	UDP	74 52690 → 33447 Len=32
17	1.371346878	172.16.21.254	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
18	1.371356795	172.16.21.1	172.16.20.1	UDP	74 58905 → 33448 Len=32
19	1.371373767	172.16.21.1	172.16.20.1	UDP	74 43796 → 33449 Len=32
20	1.371378446	172.16.21.254	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
21	1.371540620	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
22	1.371543763	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
23	1.371546137	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
24	1.371781156	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
25	1.371784718	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
26	1.371787162	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
27	1.371789677	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
28	1.371792051	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
29	1.371810978	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
30	1.371922865	172.16.21.1	172.16.2.1	DNS	86 Standard query 0x40cd PTR 254.21.16.172.in-addr.arpa
31	1.372592651	172.16.2.1	172.16.21.1	DNS	140 Standard query response 0x40cd No such name PTR 254.21.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
32	1.372669826	172.16.21.1	172.16.20.1	UDP	74 42408 → 33450 Len=32
33	1.372691408	172.16.21.1	172.16.20.1	UDP	74 34147 → 33451 Len=32
34	1.372716970	172.16.21.1	172.16.20.1	UDP	74 55341 → 33452 Len=32
35	1.372816425	172.16.21.1	172.16.2.1	DNS	86 Standard query 0x8907 PTR 253.21.16.172.in-addr.arpa
36	1.373570160	172.16.2.1	172.16.21.1	DNS	140 Standard query response 0x8907 No such name PTR 253.21.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
37	1.373669965	172.16.21.1	172.16.2.1	DNS	84 Standard query 0x3c78 PTR 1.20.16.172.in-addr.arpa
38	1.374295051	172.16.2.1	172.16.21.1	DNS	138 Standard query response 0x3c78 No such name PTR 1.20.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
39	2.002098713	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
40	3.994180676	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
41	5.996266329	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
42	6.363688766	Routerbo_eb:24:49	HewlettP_5a:76:a8	ARP	60 Who has 172.16.21.1? Tell 172.16.21.254
43	6.363707135	HewlettP_5a:76:a8	Routerbo_eb:24:49	ARP	42 172.16.21.1 is at 00:21:5a:5a:76:a8
44	7.998370210	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001

1	0.000000000	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
2	0.677276571	172.16.21.1	172.16.20.1	UDP	74 35151 → 33434 Len=32
3	0.677303251	172.16.21.1	172.16.20.1	UDP	74 54497 → 33435 Len=32
4	0.677324902	172.16.21.1	172.16.20.1	UDP	74 53617 → 33436 Len=32
5	0.677345924	172.16.21.1	172.16.20.1	UDP	74 48206 → 33437 Len=32
6	0.677367575	172.16.21.1	172.16.20.1	UDP	74 50173 → 33438 Len=32
7	0.677389086	172.16.21.1	172.16.20.1	UDP	74 45656 → 33439 Len=32
8	0.677410877	172.16.21.1	172.16.20.1	UDP	74 38192 → 33440 Len=32
9	0.677432039	172.16.21.1	172.16.20.1	UDP	74 34500 → 33441 Len=32
10	0.677452922	172.16.21.1	172.16.20.1	UDP	74 47036 → 33442 Len=32
11	0.677474154	172.16.21.1	172.16.20.1	UDP	74 51590 → 33443 Len=32
12	0.677489799	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
13	0.677501742	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
14	0.677504535	172.16.21.253	172.16.21.1	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
15	0.677526675	172.16.21.1	172.16.20.1	UDP	74 38517 → 33444 Len=32
16	0.677547418	172.16.21.1	172.16.20.1	UDP	74 36123 → 33445 Len=32
17	0.677568860	172.16.21.1	172.16.20.1	UDP	74 51086 → 33446 Len=32
18	0.677590301	172.16.21.1	172.16.20.1	UDP	74 51077 → 33447 Len=32
19	0.677611114	172.16.21.1	172.16.20.1	UDP	74 49785 → 33448 Len=32
20	0.677631857	172.16.21.1	172.16.20.1	UDP	74 57970 → 33449 Len=32
21	0.677761903	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
22	0.677767281	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
23	0.677770075	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
24	0.677772589	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
25	0.677775173	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
26	0.677794380	172.16.20.1	172.16.21.1	ICMP	102 Destination unreachable (Port unreachable)
27	0.678332932	172.16.21.1	172.16.2.1	DNS	86 Standard query 0xe2b2 PTR 253.21.16.172.in-addr.arpa
28	0.678969752	172.16.2.1	172.16.21.1	DNS	140 Standard query response 0xe2b2 No such name PTR 253.21.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
29	0.679044763	172.16.21.1	172.16.20.1	UDP	74 47259 → 33450 Len=32
30	0.679066553	172.16.21.1	172.16.20.1	UDP	74 40100 → 33451 Len=32
31	0.679087366	172.16.21.1	172.16.20.1	UDP	74 41109 → 33452 Len=32
32	0.679201698	172.16.21.1	172.16.2.1	DNS	84 Standard query 0x8719 PTR 1.20.16.172.in-addr.arpa
33	0.679908080	172.16.2.1	172.16.21.1	DNS	138 Standard query response 0x8719 No such name PTR 1.20.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
34	2.002141665	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001

1	0.000000000	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
2	2.002081323	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001
3	2.844072104	172.16.21.1	172.16.20.1	UDP	74	55533 → 33434 Len=32
4	2.844100600	172.16.21.1	172.16.20.1	UDP	74	50915 → 33435 Len=32
5	2.844122530	172.16.21.1	172.16.20.1	UDP	74	54823 → 33436 Len=32
6	2.844143623	172.16.21.1	172.16.20.1	UDP	74	60563 → 33437 Len=32
7	2.844165344	172.16.21.1	172.16.20.1	UDP	74	60846 → 33438 Len=32
8	2.844186436	172.16.21.1	172.16.20.1	UDP	74	50635 → 33439 Len=32
9	2.844207598	172.16.21.1	172.16.20.1	UDP	74	40262 → 33440 Len=32
10	2.844228830	172.16.21.1	172.16.20.1	UDP	74	40611 → 33441 Len=32
11	2.844249783	172.16.21.1	172.16.20.1	UDP	74	47339 → 33442 Len=32
12	2.844271294	172.16.21.1	172.16.20.1	UDP	74	51758 → 33443 Len=32
13	2.844288615	172.16.21.253	172.16.21.1	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
14	2.844299929	172.16.21.253	172.16.21.1	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
15	2.844302444	172.16.21.253	172.16.21.1	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
16	2.844324234	172.16.21.1	172.16.20.1	UDP	74	36999 → 33444 Len=32
17	2.844345187	172.16.21.1	172.16.20.1	UDP	74	45544 → 33445 Len=32
18	2.844365860	172.16.21.1	172.16.20.1	UDP	74	47200 → 33446 Len=32
19	2.844386324	172.16.21.1	172.16.20.1	UDP	74	41597 → 33447 Len=32
20	2.844406997	172.16.21.1	172.16.20.1	UDP	74	39577 → 33448 Len=32
21	2.844427531	172.16.21.1	172.16.20.1	UDP	74	51262 → 33449 Len=32
22	2.844516160	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
23	2.844566516	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
24	2.844571196	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
25	2.844573710	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
26	2.844576225	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
27	2.844578739	172.16.20.1	172.16.21.1	ICMP	102	Destination unreachable (Port unreachable)
28	2.845126370	172.16.21.1	172.16.2.1	DNS	86	Standard query 0x8c9a PTR 253.21.16.172.in-addr.arpa
29	2.845664364	172.16.2.1	172.16.21.1	DNS	140	Standard query response 0x8c9a No such name PTR 253.21.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
30	2.845744682	172.16.21.1	172.16.20.1	UDP	74	47668 → 33450 Len=32
31	2.845766613	172.16.21.1	172.16.20.1	UDP	74	37600 → 33451 Len=32
32	2.845786937	172.16.21.1	172.16.20.1	UDP	74	56694 → 33452 Len=32
33	2.845901338	172.16.21.1	172.16.2.1	DNS	84	Standard query 0x3060 PTR 1.20.16.172.in-addr.arpa
34	2.846601295	172.16.2.1	172.16.21.1	DNS	138	Standard query response 0x3060 No such name PTR 1.20.16.172.in-addr.arpa SOA 16.172.IN-ADDR.ARPA
35	4.003857994	Routerbo_2b:fa:05	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/74:4d:28:eb:24:49 Cost = 10 Port = 0x8001

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002089448	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	4.003883813	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
4	5.699843090	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48c3, seq=1/256, ttl=64 (reply in 5)
5	5.700396589	172.16.2.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x48c3, seq=1/256, ttl=62 (request in 4)
6	6.006056234	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
7	6.718786263	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48c3, seq=2/512, ttl=64 (reply in 8)
8	6.719259932	172.16.2.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x48c3, seq=2/512, ttl=62 (request in 7)
9	7.742789220	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48c3, seq=3/768, ttl=64 (reply in 10)
10	7.743245988	172.16.2.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x48c3, seq=3/768, ttl=62 (request in 9)
11	8.008169918	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
12	8.766785752	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48c3, seq=4/1024, ttl=64 (reply in 13)
13	8.767231345	172.16.2.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x48c3, seq=4/1024, ttl=62 (request in 12)
14	9.790786754	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48c3, seq=5/1280, ttl=64 (reply in 15)
15	9.791254766	172.16.2.254	172.16.20.1	ICMP	98	Echo (ping) reply id=0x48c3, seq=5/1280, ttl=62 (request in 14)
16	10.010308535	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
17	10.710879673	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60	Who has 172.16.20.1? Tell 172.16.20.254
18	10.710899578	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42	172.16.20.1 is at 00:21:5a:5a:78:c7
19	10.718749696	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
20	10.718864028	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60	172.16.20.254 is at 00:22:64:a7:26:a2
21	12.012465172	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002112776	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	3.507110474	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=1/256, ttl=64 (no response found!)
4	4.004201456	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
5	4.525094365	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=2/512, ttl=64 (no response found!)
6	5.549094389	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=3/768, ttl=64 (no response found!)
7	6.006274003	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
8	6.573096369	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=4/1024, ttl=64 (no response found!)
9	7.597094018	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=5/1280, ttl=64 (no response found!)
10	8.008347248	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
11	8.557062896	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42	Who has 172.16.20.254? Tell 172.16.20.1
12	8.557194688	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60	172.16.20.254 is at 00:22:64:a7:26:a2
13	8.621087687	172.16.20.1	172.16.2.254	ICMP	98	Echo (ping) request id=0x48f8, seq=6/1536, ttl=64 (no response found!)
14	10.010442284	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
15	12.012521465	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001

Experiência 5

1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002108655	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	2.628243073	172.16.20.1	172.16.2.1	DNS	73	Standard query 0xd237 A www.google.pt
4	2.628254248	172.16.20.1	172.16.2.1	DNS	73	Standard query 0xa341 AAAA www.google.pt
5	2.628911672	172.16.2.1	172.16.20.1	DNS	89	Standard query response 0xd237 A www.google.pt A 142.250.201.67
6	2.628932555	172.16.2.1	172.16.20.1	DNS	101	Standard query response 0xa341 AAAA www.google.pt AAAA 2a00:1450:4003:811::2003
7	2.629338128	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=1/256, ttl=64 (reply in 8)
8	2.699827020	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=1/256, ttl=106 (request in 7)
9	2.700024115	172.16.20.1	172.16.2.1	DNS	87	Standard query 0x597b PTR 67.201.250.142.in-addr.arpa
10	2.700660447	172.16.2.1	172.16.20.1	DNS	125	Standard query response 0x597b PTR 67.201.250.142.in-addr.arpa PTR mad07s25-in-f3.1e100.net
11	3.630707618	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=2/512, ttl=64 (reply in 12)
12	3.700269495	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=2/512, ttl=106 (request in 11)
13	4.004222129	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
14	4.632305981	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=3/768, ttl=64 (reply in 15)
15	4.702171462	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=3/768, ttl=106 (request in 14)
16	5.634189370	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=4/1024, ttl=64 (reply in 17)
17	5.703463847	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=4/1024, ttl=106 (request in 16)
18	6.006321565	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
19	6.635490485	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=5/1280, ttl=64 (reply in 20)
20	6.704852823	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=5/1280, ttl=106 (request in 19)
21	7.636893499	172.16.20.1	142.250.201.67	ICMP	98	Echo (ping) request id=0x4c2f, seq=6/1536, ttl=64 (reply in 22)
22	7.706198986	142.250.201.67	172.16.20.1	ICMP	98	Echo (ping) reply id=0x4c2f, seq=6/1536, ttl=106 (request in 21)
23	8.008444677	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
24	10.010681422	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60	RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001

Experiência 6

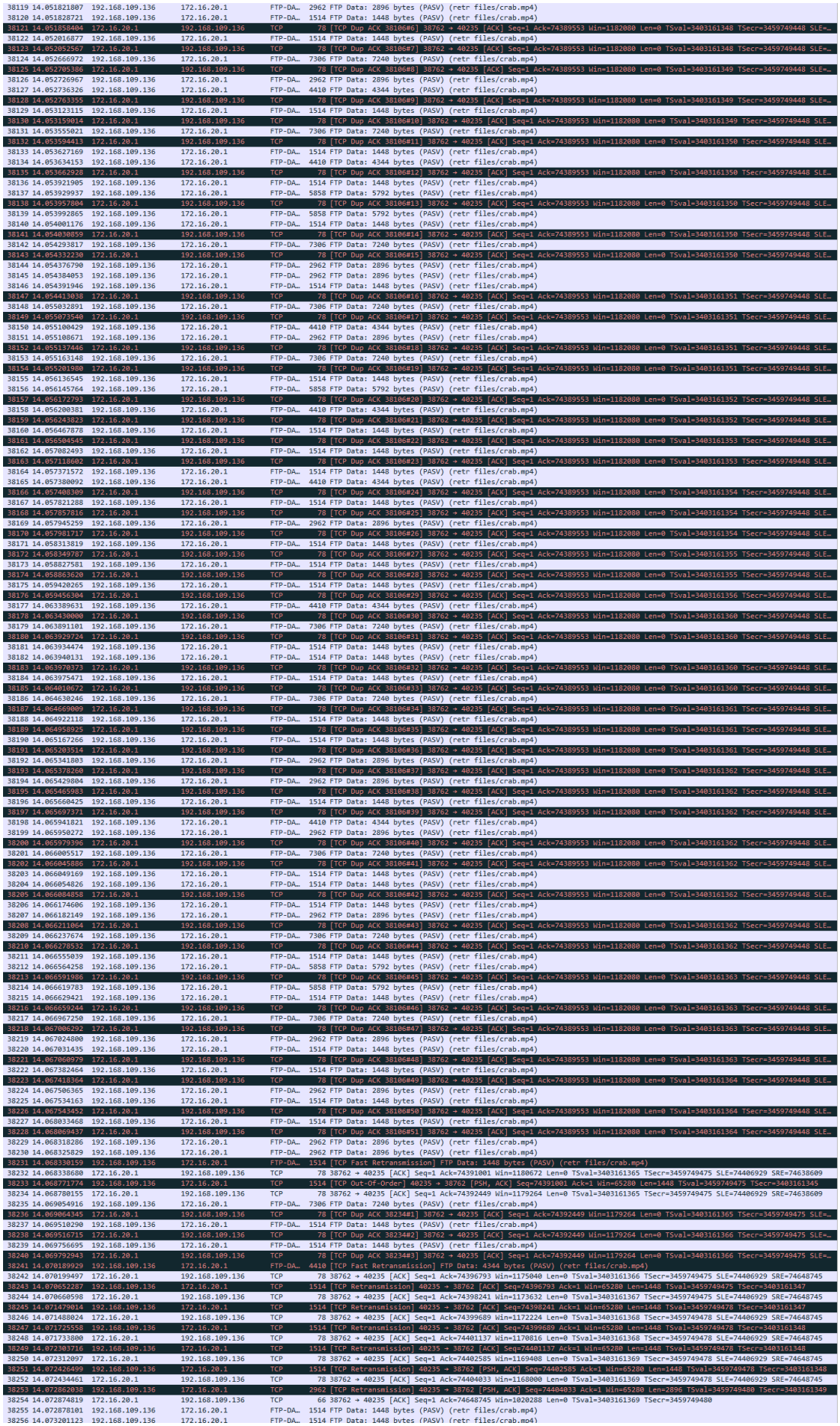
1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	2.002005236	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
3	2.965133092	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x0b8e PTR 41.163.237.44.in-addr.arpa
4	2.965861973	172.16.2.1	172.16.20.1	DNS	149 Standard query response 0x0b8e PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
5	2.966086936	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x994c PTR 25.112.82.140.in-addr.arpa
6	2.966775448	172.16.2.1	172.16.20.1	DNS	131 Standard query response 0x994c PTR 25.112.82.140.in-addr.arpa PTR lb-140-82-112-25-iad.github.com
7	3.128661144	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
8	3.128779877	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2
9	4.004195074	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
10	6.006326791	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
11	7.961218542	172.16.20.1	172.16.2.1	DNS	86 Standard query 0xf7aa PTR 41.163.237.44.in-addr.arpa
12	7.961940019	172.16.2.1	172.16.20.1	DNS	149 Standard query response 0xf7aa PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
13	7.962092696	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x9e34 PTR 25.112.82.140.in-addr.arpa
14	7.962926620	172.16.2.1	172.16.20.1	DNS	131 Standard query response 0x9e34 PTR 25.112.82.140.in-addr.arpa PTR lb-140-82-112-25-iad.github.com
15	8.008470887	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
16	8.901388587	172.16.20.1	172.16.2.1	DNS	76 Standard query 0xb8ae A netlab1.fe.up.pt
17	8.901992379	172.16.2.1	172.16.20.1	DNS	92 Standard query response 0xb8ae A netlab1.fe.up.pt A 192.168.109.136
18	8.902129620	172.16.20.1	192.168.109.136	TCP	74 45140 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3402739799 TSecr=0 WS=128
19	8.903100855	192.168.109.136	172.16.20.1	TCP	74 21 → 45140 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3459327923 TSecr=3402739799 WS=128
20	8.903120970	172.16.20.1	192.168.109.136	TCP	66 45140 → 21 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3402739800 TSecr=3459327923
21	8.905030754	192.168.109.136	172.16.20.1	FTP	100 Response: 220 Welcome to netlab-FTP server
22	8.905039833	172.16.20.1	192.168.109.136	TCP	66 45140 → 21 [ACK] Seq=1 Ack=35 Win=64256 Len=0 TSval=3402739802 TSecr=3459327925
23	8.905064069	172.16.20.1	192.168.109.136	FTP	77 Request: user rcom
24	8.905771787	192.168.109.136	172.16.20.1	TCP	66 21 → 45140 [ACK] Seq=35 Ack=12 Win=65280 Len=0 TSval=3459327926 TSecr=3402739802
25	8.905853084	192.168.109.136	172.16.20.1	FTP	100 Response: 331 Please specify the password.
26	8.905876062	172.16.20.1	192.168.109.136	FTP	77 Request: pass rcom
27	8.906416367	192.168.109.136	172.16.20.1	TCP	66 21 → 45140 [ACK] Seq=69 Ack=23 Win=65280 Len=0 TSval=3459327926 TSecr=3402739803
28	8.915625291	192.168.109.136	172.16.20.1	FTP	89 Response: 230 Login successful.
29	8.915649317	172.16.20.1	192.168.109.136	FTP	72 Request: PASV
30	8.916391188	192.168.109.136	172.16.20.1	TCP	66 21 → 45140 [ACK] Seq=92 Ack=29 Win=65280 Len=0 TSval=3459327936 TSecr=3402739813
31	8.916604419	192.168.109.136	172.16.20.1	FTP	119 Response: 227 Entering Passive Mode (192,168,109,136,187,66).
32	8.916657779	172.16.20.1	192.168.109.136	TCP	74 33544 → 47938 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3402739814 TSecr=0 WS=128
33	8.917443301	192.168.109.136	172.16.20.1	TCP	74 47938 → 33544 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3459327937 TSecr=3402739814 WS=128
34	8.917454057	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3402739815 TSecr=3459327937
35	8.917463277	172.16.20.1	192.168.109.136	FTP	87 Request: retr files/crab.mp4
36	8.918048769	192.168.109.136	172.16.20.1	TCP	66 21 → 45140 [ACK] Seq=145 Ack=50 Win=65280 Len=0 TSval=3459327938 TSecr=3402739815
37	8.918246704	192.168.109.136	172.16.20.1	FTP	144 Response: 150 Opening BINARY mode data connection for files/crab.mp4 (88123184 bytes).
38	8.918557784	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
39	8.918566235	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
40	8.918677844	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41	8.918684689	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=2897 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
42	8.918801536	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
43	8.918809149	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=4345 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
44	8.918924739	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45	8.918931584	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=5793 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
46	8.919047104	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
47	8.919053110	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=7241 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
48	8.919169887	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
49	8.919175754	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=8689 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
50	8.919292601	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
51	8.919298259	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=10137 Win=64128 Len=0 TSval=3402739816 TSecr=3459327938
52	8.919416642	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
53	8.919422300	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=11585 Win=64128 Len=0 TSval=3402739817 TSecr=3459327938
54	8.919545363	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
55	8.919550950	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=13033 Win=64128 Len=0 TSval=3402739817 TSecr=3459327938
56	8.919661581	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
57	8.919667309	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=14481 Win=64128 Len=0 TSval=3402739817 TSecr=3459327938
58	8.919784505	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
59	8.919790092	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=15929 Win=64128 Len=0 TSval=3402739817 TSecr=3459327939
60	8.919909384	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
61	8.919915181	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=17377 Win=64128 Len=0 TSval=3402739817 TSecr=3459327939
62	8.920030980	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
63	8.920036568	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=18825 Win=64128 Len=0 TSval=3402739817 TSecr=3459327939
64	8.920155580	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
65	8.920161028	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=20273 Win=64128 Len=0 TSval=3402739817 TSecr=3459327939
66	8.920289678	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
67	8.920295545	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=21721 Win=64128 Len=0 TSval=3402739817 TSecr=3459327939
68	8.920384036	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
69	8.920389624	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=23169 Win=64128 Len=0 TSval=3402739818 TSecr=3459327939
70	8.920522884	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
71	8.920659985	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
72	8.920666341	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=26065 Win=64128 Len=0 TSval=3402739818 TSecr=3459327939
73	8.920753296	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
74	8.920893540	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
75	8.920899337	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=28961 Win=64128 Len=0 TSval=3402739818 TSecr=3459327939
76	8.921015765	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
77	8.921140435	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
78	8.921145882	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=31857 Win=64128 Len=0 TSval=3402739818 TSecr=3459327940
79	8.921261333	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
80	8.921385723	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
81	8.921391240	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=34753 Win=64128 Len=0 TSval=3402739819 TSecr=3459327940
82	8.921507738	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
83	8.921633106	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
84	8.921638764	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=37649 Win=64128 Len=0 TSval=3402739819 TSecr=3459327940
85	8.921761617	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
86	8.921878534	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
87	8.921884261	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=40545 Win=64128 Len=0 TSval=3402739819 TSecr=3459327941
88	8.921999712	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
89	8.922122845	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
90	8.922128153	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=43441 Win=64128 Len=0 TSval=3402739819 TSecr=3459327941
91	8.922247095	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
92	8.922370647	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
93	8.922376095	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=46337 Win=64128 Len=0 TSval=3402739820 TSecr=3459327941
94	8.922494618	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
95	8.922616843	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
96	8.922622431	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=49233 Win=64128 Len=0 TSval=3402739820 TSecr=3459327942
97	8.922739068	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
98	8.922862411	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
99	8.922867719	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=52129 Win=64128 Len=0 TSval=3402739820 TSecr=3459327942
100	8.922984496	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
101	8.923115801	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
102	8.923121039	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=55025 Win=64128 Len=0 TSval=3402739820 TSecr=3459327942
103	8.923230972	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
104	8.923354384	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
105	8.923359832	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=57921 Win=64128 Len=0 TSval=3402739821 TSecr=3459327942
106	8.923476470	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
107	8.923599603	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
108	8.923605120	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=60817 Win=64128 Len=0 TSval=3402739821 TSecr=3459327942
109	8.923723015	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
110	8.923846078	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
111	8.923851666	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=63713 Win=64128 Len=0 TSval=3402739821 TSecr=3459327942
112	8.923968164	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
113	8.924091506	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
114	8.924096884	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=66609 Win=64128 Len=0 TSval=3402739821 TSecr=3459327942
115	8.924217014	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
116	8.924338121	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
117	8.924343569	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=69505 Win=64128 Len=0 TSval=3402739822 TSecr=3459327943
118	8.924462372	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
119	8.924585505	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
120	8.924591022	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=72401 Win=64128 Len=0 TSval=3402739822 TSecr=3459327943
121	8.924706682	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
122	8.924831003	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
123	8.924836450	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=75297 Win=64128 Len=0 TSval=3402739822 TSecr=3459327943
124	8.924953228	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
125	8.925077338	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
126	8.925083066	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=78193 Win=64128 Len=0 TSval=3402739822 TSecr=3459327944
127	8.925208503	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
128	8.925323116	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
129	8.925329192	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=81089 Win=64128 Len=0 TSval=3402739823 TSecr=3459327944
130	8.925445760	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
131	8.925568334	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
132	8.925573712	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=83985 Win=64128 Len=0 TSval=3402739823 TSecr=3459327944
133	8.925691607	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
134	8.925814810	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
135	8.925820257	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=86881 Win=64128 Len=0 TSval=3402739823 TSecr=3459327944
136	8.925940038	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
137	8.926061215	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
138	8.926066663	172.16.20.1	192.168.109.136	TCP	66 33544 → 47938 [ACK] Seq=1 Ack=89777 Win=64128 Len=0 TSval=3402739823 TSecr=3459327944

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2237 9.232263743  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3611313 Win=628608 Len=0 TSval=3402740129 TSecr=3459328240
2238 9.232307325  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2239 9.232316265  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2240 9.232330234  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3618553 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2241 9.232370114  192.168.109.136  172.16.20.1      FTP-DA… 7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2242 9.232388413  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3625793 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2243 9.232428293  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2244 9.232436884  192.168.109.136  172.16.20.1      FTP-DA… 5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2245 9.232450922  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3633033 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2246 9.232493317  192.168.109.136  172.16.20.1      FTP-DA… 7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2247 9.232511476  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3640273 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2248 9.232553452  192.168.109.136  172.16.20.1      FTP-DA… 5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2249 9.232562392  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2250 9.232575452  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3647513 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2251 9.232615402  192.168.109.136  172.16.20.1      FTP-DA… 7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2252 9.232632374  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3654753 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2253 9.232678470  192.168.109.136  172.16.20.1      FTP-DA… 7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2254 9.232696350  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3661993 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2255 9.232735881  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2256 9.232744751  192.168.109.136  172.16.20.1      FTP-DA… 5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2257 9.232760256  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3669233 Win=628608 Len=0 TSval=3402740130 TSecr=3459328240
2258 9.232800276  192.168.109.136  172.16.20.1      FTP-DA… 7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2259 9.232817737  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3676473 Win=628608 Len=0 TSval=3402740130 TSecr=3459328241
2260 9.232860062  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2261 9.232870189  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2262 9.232883808  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3683713 Win=628608 Len=0 TSval=3402740130 TSecr=3459328241
2263 9.232922292  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2264 9.232930743  192.168.109.136  172.16.20.1      FTP-DA… 2962 [TCP Previous segment not captured] FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2265 9.232939683  172.16.20.1      192.168.109.136  TCP       78 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=624512 Len=0 TSval=3402740130 TSecr=3459328241 SLE=3689505 SRE=3692401
2266 9.232981798  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2267 9.232989341  172.16.20.1      192.168.109.136  TCP       78 [TCP Window Update] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740130 TSecr=3459328241 SLE=3689…
2268 9.232994929  192.168.109.136  172.16.20.1      FTP-DA… 5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2269 9.233002891  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#1] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740130 TSecr=3459328241 SLE=368…
2270 9.233013158  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2271 9.233063863  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#2] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740130 TSecr=3459328241 SLE=368…
2272 9.233072873  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2273 9.233122881  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#3] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740130 TSecr=3459328241 SLE=368…
2274 9.233193282  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2275 9.233242941  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#4] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740130 TSecr=3459328241 SLE=368…
2276 9.233307475  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2277 9.233357273  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#5] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740131 TSecr=3459328241 SLE=368…
2278 9.233438710  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2279 9.233488578  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#6] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740131 TSecr=3459328241 SLE=368…
2280 9.233554021  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2281 9.233604378  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#7] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740131 TSecr=3459328241 SLE=368…
2282 9.234208309  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2283 9.234217179  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2284 9.234261180  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#8] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740131 TSecr=3459328241 SLE=368…
2285 9.234288698  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2286 9.234339055  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#9] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740131 TSecr=3459328241 SLE=368…
2287 9.234563949  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2288 9.234615493  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#10] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2289 9.234665780  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2290 9.234715368  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#11] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2291 9.234786329  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2292 9.234836336  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#12] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2293 9.234903944  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2294 9.234953812  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#13] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2295 9.235027636  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2296 9.235077364  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#14] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2297 9.235151397  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2298 9.235201195  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2265#15] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=36…
2299 9.235276137  192.168.109.136  172.16.20.1      FTP-DA… 1514 [TCP Previous segment not captured] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2300 9.235326284  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#16] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740132 TSecr=3459328241 SLE=37…
2301 9.235393543  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2302 9.235443830  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#17] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2303 9.235517584  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2304 9.235567521  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#18] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2305 9.235641206  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2306 9.235691283  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#19] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2307 9.235763920  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2308 9.235813857  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#20] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2309 9.235888729  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2310 9.235938667  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#21] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2311 9.236011443  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2312 9.236062777  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#22] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2313 9.236135344  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2314 9.236185491  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#23] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2315 9.236261131  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2316 9.236311348  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#24] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740133 TSecr=3459328241 SLE=37…
2317 9.236376512  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2318 9.236427078  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#25] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2319 9.236504534  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2320 9.236555519  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#26] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2321 9.236627248  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2322 9.236678023  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#27] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2323 9.236750381  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2324 9.236800877  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#28] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2325 9.236882174  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2326 9.236932112  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#29] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2327 9.236996437  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2328 9.237046305  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#30] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2329 9.237120199  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2330 9.237170066  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#31] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2331 9.237244030  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2332 9.237293968  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#32] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740134 TSecr=3459328241 SLE=37…
2333 9.237366604  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2334 9.237416961  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2265#33] 33544 → 47938 [ACK] Seq=1 Ack=3688057 Win=626560 Len=0 TSval=3402740135 TSecr=3459328241 SLE=37…
2335 9.237488340  192.168.109.136  172.16.20.1      FTP-DA… 1514 [TCP Fast Retransmission] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2336 9.237497769  172.16.20.1      192.168.109.136  TCP       78 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=587520 Len=0 TSval=3402740135 TSecr=3459328254 SLE=3728601 SRE=3754665
2337 9.237610496  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2338 9.237618178  172.16.20.1      192.168.109.136  TCP       78 [TCP Window Update] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740135 TSecr=3459328254 SLE=3728…
2339 9.237733978  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2340 9.237740334  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#1] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740135 TSecr=3459328254 SLE=372…
2341 9.237853479  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2342 9.237903626  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#2] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740135 TSecr=3459328254 SLE=372…
2343 9.237978079  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2344 9.238028086  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#3] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740135 TSecr=3459328254 SLE=372…
2345 9.238100793  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2346 9.238150521  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#4] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740135 TSecr=3459328254 SLE=372…
2347 9.238840499  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2348 9.238890436  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#5] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740136 TSecr=3459328254 SLE=372…
2349 9.239701452  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2350 9.239751250  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#6] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740137 TSecr=3459328254 SLE=372…
2351 9.239837925  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2352 9.239843652  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2353 9.239888009  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2336#7] 33544 → 47938 [ACK] Seq=1 Ack=3727153 Win=610560 Len=0 TSval=3402740137 TSecr=3459328254 SLE=372…
2354 9.240110801  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2355 9.240118135  192.168.109.136  172.16.20.1      FTP-DA… 1514 [TCP Fast Retransmission] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2356 9.240126655  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3772041 Win=598528 Len=0 TSval=3402740137 TSecr=3459328256
2357 9.240129170  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2358 9.240182041  192.168.109.136  172.16.20.1      FTP-DA… 4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2359 9.240190003  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2360 9.240213610  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3780729 Win=611712 Len=0 TSval=3402740137 TSecr=3459328256
2361 9.240236099  192.168.109.136  172.16.20.1      FTP-DA… 2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2362 9.240241826  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2363 9.240256843  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3785073 Win=608640 Len=0 TSval=3402740137 TSecr=3459328257
2364 9.240335486  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2365 9.240448561  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2366 9.240455266  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3787969 Win=628608 Len=0 TSval=3402740138 TSecr=3459328257
2367 9.240561916  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2368 9.240685468  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2369 9.240692173  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3790865 Win=628608 Len=0 TSval=3402740138 TSecr=3459328257
2370 9.240809509  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2371 9.241671510  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2372 9.241678006  172.16.20.1      192.168.109.136  TCP       66 33544 → 47938 [ACK] Seq=1 Ack=3793761 Win=628608 Len=0 TSval=3402740139 TSecr=3459328258
2373 9.242900815  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2374 9.244132075  192.168.109.136  172.16.20.1      FTP-DA… 1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
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41515 16.392449159  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87908081 Win=713600 Len=0 TSval=3402747290 TSecr=3459335405
41516 16.392477236  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41517 16.392495186  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87915321 Win=713600 Len=0 TSval=3402747290 TSecr=3459335405
41518 16.392535695  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41519 16.392544634  192.168.109.136   172.16.20.1       FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
41520 16.392566286  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87922561 Win=713600 Len=0 TSval=3402747290 TSecr=3459335405
41521 16.392599810  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41522 16.392618249  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87929801 Win=713600 Len=0 TSval=3402747290 TSecr=3459335405
41523 16.392627747  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41524 16.392739845  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41525 16.392746061  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87932697 Win=713600 Len=0 TSval=3402747290 TSecr=3459335405
41526 16.392856553  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41527 16.392980873  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41528 16.392986810  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87935593 Win=713600 Len=0 TSval=3402747290 TSecr=3459335406
41529 16.393103727  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41530 16.393225253  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41531 16.393231120  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87938489 Win=713600 Len=0 TSval=3402747290 TSecr=3459335406
41532 16.393472218  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41533 16.394703059  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41534 16.394708926  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87941385 Win=713600 Len=0 TSval=3402747292 TSecr=3459335406
41535 16.395932992  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41536 16.396208662  192.168.109.136   172.16.20.1       FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
41537 16.396215925  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41538 16.396223399  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87945729 Win=713600 Len=0 TSval=3402747293 TSecr=3459335406
41539 16.397164811  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41540 16.397171097  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87948625 Win=713600 Len=0 TSval=3402747294 TSecr=3459335406
41541 16.398270773  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41542 16.399502662  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41543 16.399508459  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87951521 Win=713600 Len=0 TSval=3402747297 TSecr=3459335406
41544 16.400034096  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41545 16.400051766  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87958761 Win=713600 Len=0 TSval=3402747297 TSecr=3459335406
41546 16.400103729  192.168.109.136   172.16.20.1       FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
41547 16.400112320  192.168.109.136   172.16.20.1       FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
41548 16.400118885  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87961657 Win=711680 Len=0 TSval=3402747297 TSecr=3459335406
41549 16.400141584  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87966001 Win=713600 Len=0 TSval=3402747297 TSecr=3459335413
41550 16.400166448  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41551 16.400185934  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87973241 Win=713600 Len=0 TSval=3402747297 TSecr=3459335413
41552 16.400227910  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41553 16.400251726  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87980481 Win=713600 Len=0 TSval=3402747297 TSecr=3459335413
41554 16.400289581  192.168.109.136   172.16.20.1       FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
41555 16.400299988  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41556 16.400309975  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87987721 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41557 16.400352370  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41558 16.400372555  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87994961 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41559 16.400410689  192.168.109.136   172.16.20.1       FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
41560 16.400419070  192.168.109.136   172.16.20.1       FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
41561 16.400426124  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=87997857 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41562 16.400447077  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88002201 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41563 16.400474595  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41564 16.400492335  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88009441 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41565 16.400532774  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41566 16.400541435  192.168.109.136   172.16.20.1       FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
41567 16.400564623  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88016681 Win=713600 Len=0 TSval=3402747298 TSecr=3459335413
41568 16.400596541  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41569 16.400614141  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88023921 Win=713600 Len=0 TSval=3402747298 TSecr=3459335414
41570 16.400625526  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41571 16.400737903  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41572 16.400744049  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88026817 Win=713600 Len=0 TSval=3402747298 TSecr=3459335414
41573 16.400854191  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41574 16.400978302  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41575 16.400985077  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88029713 Win=713600 Len=0 TSval=3402747298 TSecr=3459335414
41576 16.401101854  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41577 16.401470415  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41578 16.401476142  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88032609 Win=713600 Len=0 TSval=3402747299 TSecr=3459335414
41579 16.402701186  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41580 16.403085252  192.168.109.136   172.16.20.1       FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
41581 16.403099710  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88036953 Win=713600 Len=0 TSval=3402747300 TSecr=3459335414
41582 16.403684783  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41583 16.404915345  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41584 16.404921561  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88039849 Win=713600 Len=0 TSval=3402747302 TSecr=3459335414
41585 16.405653654  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41586 16.405807239  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41587 16.405824210  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88048537 Win=713600 Len=0 TSval=3402747303 TSecr=3459335414
41588 16.405870377  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41589 16.405889025  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88055777 Win=713600 Len=0 TSval=3402747303 TSecr=3459335414
41590 16.405930861  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41591 16.405948880  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88063017 Win=713600 Len=0 TSval=3402747303 TSecr=3459335421
41592 16.405992322  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41593 16.406010132  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88070257 Win=713600 Len=0 TSval=3402747303 TSecr=3459335421
41594 16.406051549  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41595 16.406059790  192.168.109.136   172.16.20.1       FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
41596 16.406065867  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41597 16.406080324  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88077497 Win=713600 Len=0 TSval=3402747303 TSecr=3459335421
41598 16.406115665  192.168.109.136   172.16.20.1       FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
41599 16.406134243  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88084737 Win=713600 Len=0 TSval=3402747303 TSecr=3459335421
41600 16.406183133  192.168.109.136   172.16.20.1       FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
41601 16.406199895  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88090529 Win=713600 Len=0 TSval=3402747303 TSecr=3459335421
41602 16.406272462  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41603 16.406394547  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41604 16.406400554  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88093425 Win=713600 Len=0 TSval=3402747304 TSecr=3459335421
41605 16.406512652  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41606 16.406637880  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41607 16.406643747  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88096321 Win=713600 Len=0 TSval=3402747304 TSecr=3459335421
41608 16.406759057  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41609 16.406883029  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41610 16.406889035  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88099217 Win=713600 Len=0 TSval=3402747304 TSecr=3459335421
41611 16.407005533  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41612 16.407128806  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41613 16.407134463  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88102113 Win=713600 Len=0 TSval=3402747304 TSecr=3459335421
41614 16.407252008  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41615 16.407375211  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41616 16.407380938  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88105009 Win=713600 Len=0 TSval=3402747305 TSecr=3459335421
41617 16.407498484  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41618 16.407620430  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41619 16.407626296  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88107905 Win=713600 Len=0 TSval=3402747305 TSecr=3459335421
41620 16.407743912  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41621 16.407869210  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41622 16.407875007  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88110801 Win=713600 Len=0 TSval=3402747305 TSecr=3459335421
41623 16.407992413  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41624 16.408112682  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41625 16.408118479  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88113697 Win=713600 Len=0 TSval=3402747305 TSecr=3459335421
41626 16.408237911  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41627 16.408359088  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41628 16.408364675  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88116593 Win=713600 Len=0 TSval=3402747306 TSecr=3459335421
41629 16.408482361  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41630 16.408605563  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41631 16.408611221  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88119489 Win=713600 Len=0 TSval=3402747306 TSecr=3459335421
41632 16.408727858  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41633 16.408852877  192.168.109.136   172.16.20.1       FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
41634 16.408858814  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88122385 Win=713600 Len=0 TSval=3402747306 TSecr=3459335421
41635 16.408891151  192.168.109.136   172.16.20.1       FTP-DA…   866 FTP Data: 800 bytes (PASV) (retr files/crab.mp4)
41636 16.452650369  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [ACK] Seq=1 Ack=88123186 Win=713600 Len=0 TSval=3402747350 TSecr=3459335421
41637 16.453481709  192.168.109.136   172.16.20.1       FTP        90 Response: 226 Transfer complete.
41638 16.453490718  172.16.20.1       192.168.109.136   TCP        66 45140 → 21 [ACK] Seq=50 Ack=247 Win=64256 Len=0 TSval=3402747351 TSecr=3459335473
41639 16.453542193  172.16.20.1       192.168.109.136   FTP        72 Request: quit
41640 16.454728544  192.168.109.136   172.16.20.1       TCP        66 21 → 45140 [ACK] Seq=247 Ack=56 Win=65280 Len=0 TSval=3459335474 TSecr=3402747351
41641 16.454773173  192.168.109.136   172.16.20.1       FTP        80 Response: 221 Goodbye.
41642 16.454797549  172.16.20.1       192.168.109.136   TCP        66 45140 → 21 [FIN, ACK] Seq=56 Ack=261 Win=64256 Len=0 TSval=3402747352 TSecr=3459335474
41643 16.454810400  192.168.109.136   172.16.20.1       TCP        66 21 → 45140 [FIN, ACK] Seq=261 Ack=56 Win=65280 Len=0 TSval=3459335474 TSecr=3402747351
41644 16.454812844  172.16.20.1       192.168.109.136   TCP        66 33544 → 47938 [FIN, ACK] Seq=1 Ack=88123186 Win=713600 Len=0 TSval=3402747352 TSecr=3459335421
41645 16.454817733  172.16.20.1       192.168.109.136   TCP        66 45140 → 21 [ACK] Seq=57 Ack=262 Win=64256 Len=0 TSval=3402747352 TSecr=3459335474
41646 16.455397778  192.168.109.136   172.16.20.1       TCP        66 21 → 45140 [ACK] Seq=262 Ack=57 Win=65280 Len=0 TSval=3459335475 TSecr=3402747352
41647 16.455434935  192.168.109.136   172.16.20.1       TCP        66 47938 → 33544 [ACK] Seq=88123186 Ack=2 Win=65280 Len=0 TSval=3459335475 TSecr=3402747352
41648 17.965158393  172.16.20.1       172.16.2.1        DNS        86 Standard query 0x0ddd PTR 41.163.237.44.in-addr.arpa
41649 17.966108047  172.16.2.1        172.16.20.1       DNS       149 Standard query response 0x0ddd PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
41650 17.966269105  172.16.20.1       172.16.2.1        DNS        86 Standard query 0xf96a PTR 25.112.82.140.in-addr.arpa
41651 17.967123004  172.16.2.1        172.16.20.1       DNS       131 Standard query response 0xf96a PTR 25.112.82.140.in-addr.arpa PTR lb-140-82-112-25-iad.github.com
41652 18.019129757  Routerbo_2b:fa:06  Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
```

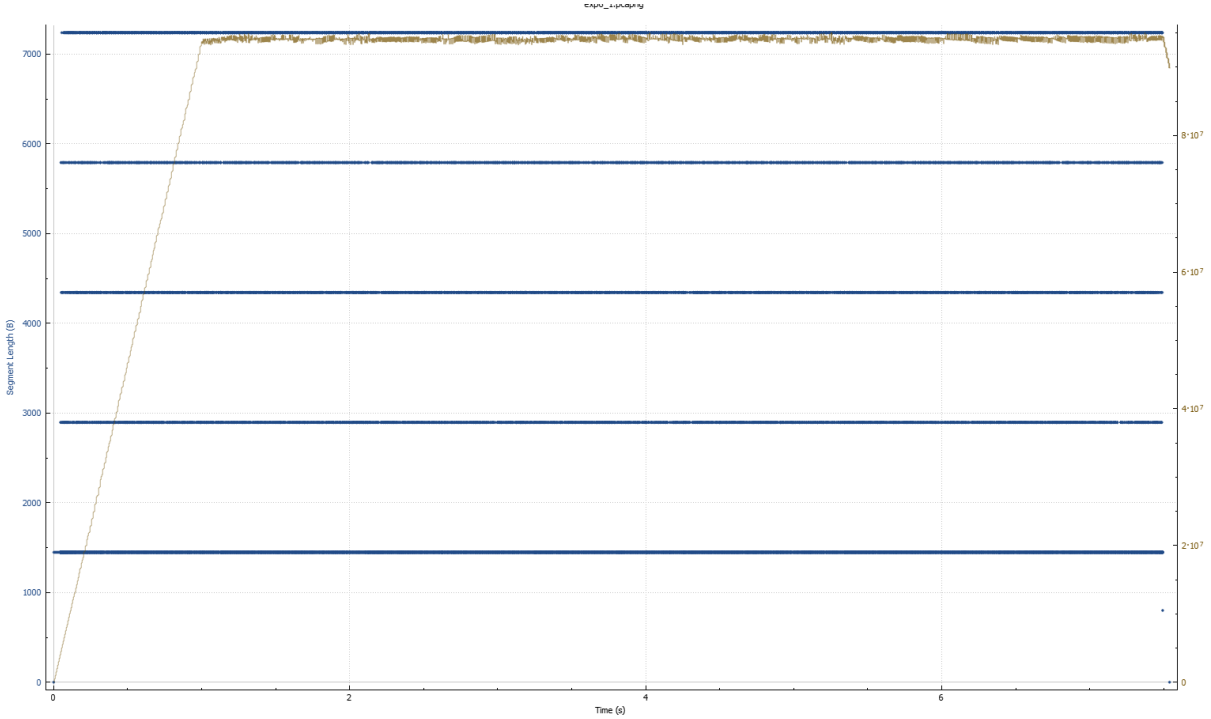
1	0.000000000	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
2	1.566267178	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x11ca PTR 26.114.82.140.in-addr.arpa
3	1.566923140	172.16.2.1	172.16.20.1	DNS	131 Standard query response 0x11ca PTR 26.114.82.140.in-addr.arpa PTR lb-140-82-114-26-iad.github.com
4	1.567074699	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x8dea PTR 41.163.237.44.in-addr.arpa
5	1.567596982	172.16.2.1	172.16.20.1	DNS	149 Standard query response 0x8dea PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
6	2.002199670	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
7	4.004391703	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
8	6.006558151	Routerbo_2b:fa:06	Spanning-tree-(for-…	STP	60 RST. Root = 32768/0/c4:ad:34:2b:fa:06 Cost = 0 Port = 0x8001
9	6.566213146	172.16.20.1	172.16.2.1	DNS	86 Standard query 0xd5f2 PTR 26.114.82.140.in-addr.arpa
10	6.567181584	172.16.2.1	172.16.20.1	DNS	131 Standard query response 0xd5f2 PTR 26.114.82.140.in-addr.arpa PTR lb-140-82-114-26-iad.github.com
11	6.567410738	172.16.20.1	172.16.2.1	DNS	86 Standard query 0x3116 PTR 41.163.237.44.in-addr.arpa
12	6.568079551	172.16.2.1	172.16.20.1	DNS	149 Standard query response 0x3116 PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
13	6.573657742	HewlettP_5a:78:c7	HewlettP_a7:26:a2	ARP	42 Who has 172.16.20.254? Tell 172.16.20.1
14	6.573765579	HewlettP_a7:26:a2	HewlettP_5a:78:c7	ARP	60 172.16.20.254 is at 00:22:64:a7:26:a2
15	6.598121100	172.16.20.1	172.16.2.1	DNS	76 Standard query 0xe4d9 A netlab1.fe.up.pt
16	6.598778948	172.16.2.1	172.16.20.1	DNS	92 Standard query response 0xe4d9 A netlab1.fe.up.pt A 192.168.109.136
17	6.598968012	172.16.20.1	192.168.109.136	TCP	74 45878 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3403153895 TSecr=0 WS=128
18	6.599760586	192.168.109.136	172.16.20.1	TCP	74 21 → 45878 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3459742016 TSecr=3403153895 WS=128
19	6.599778745	172.16.20.1	192.168.109.136	TCP	66 45878 → 21 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3403153896 TSecr=3459742016
20	6.616828099	192.168.109.136	172.16.20.1	FTP	100 Response: 220 Welcome to netlab-FTP server
21	6.616852544	172.16.20.1	192.168.109.136	TCP	66 45878 → 21 [ACK] Seq=1 Ack=35 Win=64256 Len=0 TSval=3403153913 TSecr=3459742033
22	6.616902621	172.16.20.1	192.168.109.136	FTP	77 Request: user rcom
23	6.617346052	192.168.109.136	172.16.20.1	TCP	66 21 → 45878 [ACK] Seq=35 Ack=12 Win=65280 Len=0 TSval=3459742034 TSecr=3403153913
24	6.617408212	192.168.109.136	172.16.20.1	FTP	100 Response: 331 Please specify the password.
25	6.617431051	172.16.20.1	192.168.109.136	FTP	77 Request: pass rcom
26	6.617941461	192.168.109.136	172.16.20.1	TCP	66 21 → 45878 [ACK] Seq=69 Ack=23 Win=65280 Len=0 TSval=3459742034 TSecr=3403153914
27	6.626840948	192.168.109.136	172.16.20.1	FTP	89 Response: 230 Login successful.
28	6.626900035	172.16.20.1	192.168.109.136	FTP	72 Request: PASV
29	6.627460592	192.168.109.136	172.16.20.1	TCP	66 21 → 45878 [ACK] Seq=92 Ack=29 Win=65280 Len=0 TSval=3459742044 TSecr=3403153923
30	6.627554949	192.168.109.136	172.16.20.1	FTP	119 Response: 227 Entering Passive Mode (192,168,109,136,157,43).
31	6.627639738	172.16.20.1	192.168.109.136	TCP	74 38762 → 40235 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3403153924 TSecr=0 WS=128
32	6.628148542	192.168.109.136	172.16.20.1	TCP	74 40235 → 38762 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3459742045 TSecr=3403153924 WS=128
33	6.628167399	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3403153924 TSecr=3459742045
34	6.628180181	172.16.20.1	192.168.109.136	FTP	87 Request: retr files/crab.mp4
35	6.628635066	192.168.109.136	172.16.20.1	TCP	66 21 → 45878 [ACK] Seq=145 Ack=50 Win=65280 Len=0 TSval=3459742045 TSecr=3403153924
36	6.628766370	192.168.109.136	172.16.20.1	FTP	144 Response: 150 Opening BINARY mode data connection for files/crab.mp4 (88123184 bytes).
37	6.629066972	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
38	6.629084084	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 TSval=3403153925 TSecr=3459742045
39	6.629174181	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40	6.629181026	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=2897 Win=64128 Len=0 TSval=3403153925 TSecr=3459742045
41	6.629308139	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
42	6.629314425	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=4345 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
43	6.629431411	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
44	6.629437278	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=5793 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
45	6.629554265	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
46	6.629560131	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=7241 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
47	6.629677327	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
48	6.629683753	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=8689 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
49	6.629802415	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
50	6.629808422	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=10137 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
51	6.629924361	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
52	6.629930227	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=11585 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
53	6.630046096	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
54	6.630055595	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=13033 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
55	6.630169578	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
56	6.630175724	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=14481 Win=64128 Len=0 TSval=3403153926 TSecr=3459742045
57	6.630291803	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
58	6.630297669	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=15929 Win=64128 Len=0 TSval=3403153926 TSecr=3459742046
59	6.630415215	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
60	6.630420942	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=17377 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
61	6.630538068	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
62	6.630543935	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=18825 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
63	6.630667696	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
64	6.630673493	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=20273 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
65	6.630784403	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
66	6.630790549	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=21721 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
67	6.630908304	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
68	6.630914240	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=23169 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
69	6.631030877	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
70	6.631154150	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
71	6.631160435	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=26065 Win=64128 Len=0 TSval=3403153927 TSecr=3459742046
72	6.631277631	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
73	6.631400415	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
74	6.631406282	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=28961 Win=64128 Len=0 TSval=3403153928 TSecr=3459742046
75	6.631523128	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
76	6.631646051	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
77	6.631651639	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=31857 Win=64128 Len=0 TSval=3403153928 TSecr=3459742047
78	6.631769463	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
79	6.631892456	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
80	6.631898602	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=34753 Win=64128 Len=0 TSval=3403153928 TSecr=3459742047
81	6.632015868	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
82	6.632147102	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
83	6.632153109	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=37649 Win=64128 Len=0 TSval=3403153928 TSecr=3459742047
84	6.632262273	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
85	6.632385475	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
86	6.632391342	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=40545 Win=64128 Len=0 TSval=3403153929 TSecr=3459742048
87	6.632508747	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
88	6.632629785	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
89	6.632635372	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=43441 Win=64128 Len=0 TSval=3403153929 TSecr=3459742048
90	6.632754105	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
91	6.632879263	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
92	6.632885129	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=46337 Win=64128 Len=0 TSval=3403153929 TSecr=3459742048
93	6.633000579	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
94	6.633122804	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
95	6.633128461	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=49233 Win=64128 Len=0 TSval=3403153929 TSecr=3459742049
96	6.633246076	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
97	6.633369209	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
98	6.633374796	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=52129 Win=64128 Len=0 TSval=3403153930 TSecr=3459742049
99	6.633492690	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
100	6.633613378	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
101	6.633618966	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=55025 Win=64128 Len=0 TSval=3403153930 TSecr=3459742049
102	6.633740143	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
103	6.633861180	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
104	6.633867536	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=57921 Win=64128 Len=0 TSval=3403153930 TSecr=3459742049
105	6.633983475	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
106	6.634107585	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
107	6.634113312	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=60817 Win=64128 Len=0 TSval=3403153930 TSecr=3459742049
108	6.634230927	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
109	6.634354060	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
110	6.634359647	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=63713 Win=64128 Len=0 TSval=3403153931 TSecr=3459742049
111	6.634476075	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
112	6.634600394	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
113	6.634607449	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=66609 Win=64128 Len=0 TSval=3403153931 TSecr=3459742049
114	6.634721851	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
115	6.634846310	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
116	6.634852457	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=69505 Win=64128 Len=0 TSval=3403153931 TSecr=3459742050
117	6.634971329	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
118	6.635091458	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
119	6.635097045	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=72401 Win=64128 Len=0 TSval=3403153931 TSecr=3459742050
120	6.635214591	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
121	6.635337933	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
122	6.635343730	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=75297 Win=64128 Len=0 TSval=3403153932 TSecr=3459742050
123	6.635460227	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
124	6.635583150	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
125	6.635589087	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=78193 Win=64128 Len=0 TSval=3403153932 TSecr=3459742051
126	6.635718016	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
127	6.635830044	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
128	6.635835841	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=81089 Win=64128 Len=0 TSval=3403153932 TSecr=3459742051
129	6.635951989	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
130	6.636075331	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
131	6.636080919	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=83985 Win=64128 Len=0 TSval=3403153932 TSecr=3459742051
132	6.636199651	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
133	6.636321736	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
134	6.636327323	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=86881 Win=64128 Len=0 TSval=3403153933 TSecr=3459742051
135	6.636446056	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
136	6.636568420	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
137	6.636574217	172.16.20.1	192.168.109.136	TCP	66 38762 → 40235 [ACK] Seq=1 Ack=89777 Win=64128 Len=0 TSval=3403153933 TSecr=3459742051
138	6.636694137	192.168.109.136	172.16.20.1	FTP-DA…	1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)

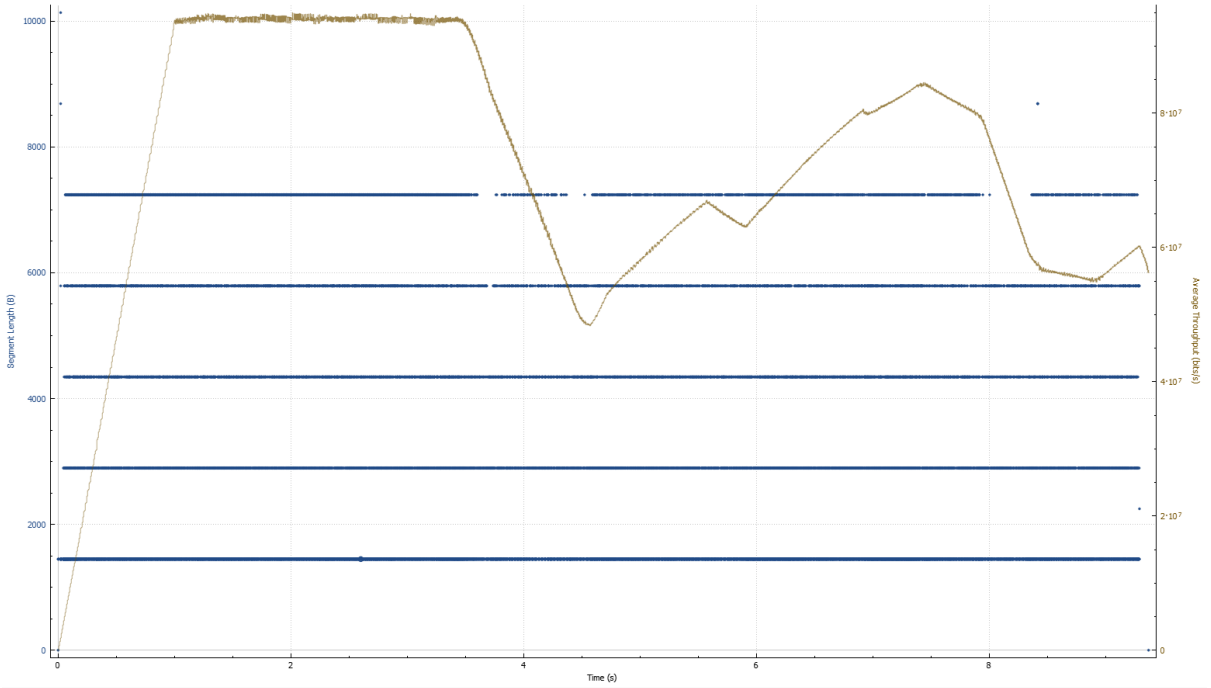
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2273 6.956536292  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3787969 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742361
2274 6.956580503  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2275 6.956597475  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3795209 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742361
2276 6.956642034  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2277 6.956658098  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3802449 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742361
2278 6.956704334  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2279 6.956720048  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3809689 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742362
2280 6.956764887  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2281 6.956780672  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3816929 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742362
2282 6.956827816  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2283 6.956843390  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3824169 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742362
2284 6.956885785  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2285 6.956893537  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2286 6.956910369  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3831409 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742362
2287 6.956950529  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2288 6.956966802  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3838649 Win=1188096 Len=0 TSval=3403154253 TSecr=3459742362
2289 6.957008848  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2290 6.957015972  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2291 6.957020791  192.168.109.136  172.16.20.1      FTP-DA…  2962 [TCP Previous segment not captured] FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2292 6.957028892  172.16.20.1      192.168.109.136  TCP       78 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154253 TSecr=3459742362 SLE=3844441 SRE=3847337
2293 6.957072544  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2294 6.957081205  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#1] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154253 TSecr=3459742362 SLE=38…
2295 6.957101110  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2296 6.957106697  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#2] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154253 TSecr=3459742362 SLE=38…
2297 6.957166413  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2298 6.957202591  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#3] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154253 TSecr=3459742362 SLE=38…
2299 6.957273062  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2300 6.957308403  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#4] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154253 TSecr=3459742362 SLE=38…
2301 6.957394658  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2302 6.957429649  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#5] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154254 TSecr=3459742362 SLE=38…
2303 6.957762310  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2304 6.957797360  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#6] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154254 TSecr=3459742362 SLE=38…
2305 6.959040360  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2306 6.959082685  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#7] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154255 TSecr=3459742362 SLE=38…
2307 6.959161118  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2308 6.959169499  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2309 6.959198484  172.16.20.1      192.168.109.136  TCP       78 [TCP Dup ACK 2292#8] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154255 TSecr=3459742362 SLE=38…
2310 6.959229005  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2311 6.959237526  192.168.109.136  172.16.20.1      FTP-DA…  1514 [TCP Previous segment not captured] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2312 6.959269304  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2292#9] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154255 TSecr=3459742362 SLE=38…
2313 6.959358144  192.168.109.136  172.16.20.1      FTP-DA…  1514 [TCP Previous segment not captured] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2314 6.959394532  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#10] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2315 6.959482045  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2316 6.959519131  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#11] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2317 6.959602873  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2318 6.959640448  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#12] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2319 6.959727542  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2320 6.959764908  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#13] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2321 6.959853398  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2322 6.959890485  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#14] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2323 6.959977020  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2324 6.960014176  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#15] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2325 6.960107206  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2326 6.960144502  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#16] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2327 6.960224751  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2328 6.960262466  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#17] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154256 TSecr=3459742362 SLE=3…
2329 6.960353052  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2330 6.960390069  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#18] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2331 6.960474159  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2332 6.960511665  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#19] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2333 6.960595895  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2334 6.960635845  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#20] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2335 6.960718679  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2336 6.960756813  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#21] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2337 6.960842859  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2338 6.960880644  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#22] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2339 6.960966829  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2340 6.961002519  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#23] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2341 6.961089194  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2342 6.961125931  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#24] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2343 6.961207018  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2344 6.961248644  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#25] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154257 TSecr=3459742362 SLE=3…
2345 6.961333852  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2346 6.961371218  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#26] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2347 6.961457683  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2348 6.961494141  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#27] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2349 6.961582352  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2350 6.961618461  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#28] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2351 6.961702202  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2352 6.961740197  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#29] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2353 6.961824776  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2354 6.961864656  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#30] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2355 6.961941274  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2356 6.961978500  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#31] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2357 6.962066432  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2358 6.962103379  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#32] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2359 6.962190821  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2360 6.962226930  172.16.20.1      192.168.109.136  TCP       94 [TCP Dup ACK 2292#33] 38762 → 40235 [ACK] Seq=1 Ack=3842993 Win=1185024 Len=0 TSval=3403154258 TSecr=3459742362 SLE=3…
2361 6.962453220  192.168.109.136  172.16.20.1      FTP-DA…  1514 [TCP Fast Retransmission] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2362 6.962469843  172.16.20.1      192.168.109.136  TCP       86 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1147392 Len=0 TSval=3403154259 TSecr=3459742374 SLE=3890777 SRE=3925529 SLE…
2363 6.962473056  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2364 6.962483183  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2362#1] 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1147392 Len=0 TSval=3403154259 TSecr=3459742374 SLE=38…
2365 6.962566924  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2366 6.962583407  172.16.20.1      192.168.109.136  TCP       86 [TCP Window Update] 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1152896 Len=0 TSval=3403154259 TSecr=3459742374 SLE=389…
2367 6.962671269  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2368 6.962708286  172.16.20.1      192.168.109.136  TCP       86 [TCP Window Update] 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1161088 Len=0 TSval=3403154259 TSecr=3459742374 SLE=389…
2369 6.962795030  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2370 6.962831209  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2362#2] 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1161088 Len=0 TSval=3403154259 TSecr=3459742374 SLE=38…
2371 6.963104922  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2372 6.963111906  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2373 6.963141310  172.16.20.1      192.168.109.136  TCP       86 [TCP Dup ACK 2362#3] 38762 → 40235 [ACK] Seq=1 Ack=3880641 Win=1161088 Len=0 TSval=3403154259 TSecr=3459742374 SLE=38…
2374 6.963189152  192.168.109.136  172.16.20.1      FTP-DA…  1514 [TCP Fast Retransmission] FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2375 6.963197324  172.16.20.1      192.168.109.136  TCP       78 38762 → 40235 [ACK] Seq=1 Ack=3883537 Win=1158272 Len=0 TSval=3403154259 TSecr=3459742376 SLE=3890777 SRE=3934217
2376 6.964283376  192.168.109.136  172.16.20.1      TCP     1514 [TCP Retransmission] 40235 → 38762 [ACK] Seq=3883537 Ack=1 Win=65280 Len=1448 TSval=3459742377 TSecr=3403154256
2377 6.964290919  172.16.20.1      192.168.109.136  TCP       78 38762 → 40235 [ACK] Seq=1 Ack=3884985 Win=1158016 Len=0 TSval=3403154260 TSecr=3459742377 SLE=3890777 SRE=3934217
2378 6.964776396  192.168.109.136  172.16.20.1      TCP     1514 [TCP Retransmission] 40235 → 38762 [ACK] Seq=3884985 Ack=1 Win=65280 Len=1448 TSval=3459742377 TSecr=3403154256
2379 6.964783939  172.16.20.1      192.168.109.136  TCP       78 38762 → 40235 [ACK] Seq=1 Ack=3886433 Win=1158016 Len=0 TSval=3403154261 TSecr=3459742377 SLE=3890777 SRE=3934217
2380 6.964938788  192.168.109.136  172.16.20.1      TCP     4410 [TCP Retransmission] 40235 → 38762 [PSH, ACK] Seq=3886433 Ack=1 Win=65280 Len=4344 TSval=3459742377 TSecr=3403154256
2381 6.964948069  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3934217 Win=1156096 Len=0 TSval=3403154261 TSecr=3459742377
2382 6.964952609  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
2383 6.964958336  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3937113 Win=1157248 Len=0 TSval=3403154261 TSecr=3459742377
2384 6.965008832  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2385 6.965015816  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2386 6.965022731  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3944313 Win=1170048 Len=0 TSval=3403154261 TSecr=3459742377
2387 6.965066312  192.168.109.136  172.16.20.1      FTP-DA…  4344 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
2388 6.965073716  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3948697 Win=1186048 Len=0 TSval=3403154261 TSecr=3459742378
2389 6.965155082  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2390 6.965269834  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2391 6.965275282  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3951593 Win=1188096 Len=0 TSval=3403154261 TSecr=3459742378
2392 6.965396179  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2393 6.966252869  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2394 6.966258666  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3954489 Win=1188096 Len=0 TSval=3403154262 TSecr=3459742379
2395 6.967359944  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2396 6.968590502  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2397 6.968595949  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3957385 Win=1188096 Len=0 TSval=3403154265 TSecr=3459742379
2398 6.969822456  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2399 6.970834964  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2400 6.970840901  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3960281 Win=1188096 Len=0 TSval=3403154267 TSecr=3459742379
2401 6.970845860  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2402 6.970849003  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2403 6.970856476  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3963177 Win=1188096 Len=0 TSval=3403154267 TSecr=3459742380
2404 6.971050638  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2405 6.972282872  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
2406 6.972289018  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3966073 Win=1188096 Len=0 TSval=3403154268 TSecr=3459742380
2407 6.973553868  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
2408 6.973571678  172.16.20.1      192.168.109.136  TCP       66 38762 → 40235 [ACK] Seq=1 Ack=3973313 Win=1188096 Len=0 TSval=3403154270 TSecr=3459742380
2409 6.973623082  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
2410 6.973631952  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
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40300 14.521440621  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40301 14.521477009  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#4] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161818 TSecr=3459749917 SLE=…
40302 14.522179137  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40303 14.522216293  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#5] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161818 TSecr=3459749917 SLE=…
40304 14.522795149  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40305 14.522838839  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#6] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161819 TSecr=3459749917 SLE=…
40306 14.523326442  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
40307 14.523365975  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#7] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161820 TSecr=3459749917 SLE=…
40308 14.523386157  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40309 14.523392792  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40310 14.523422336  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#8] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161820 TSecr=3459749917 SLE=…
40311 14.524053853  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40312 14.524090381  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#9] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161820 TSecr=3459749917 SLE=…
40313 14.524402158  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40314 14.524437149  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#10] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161821 TSecr=3459749917 SLE…
40315 14.525017472  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40316 14.525052602  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#11] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161821 TSecr=3459749917 SLE…
40317 14.526882339  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40318 14.526919077  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#12] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161823 TSecr=3459749917 SLE…
40319 14.528836257  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40320 14.528874950  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#13] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161825 TSecr=3459749917 SLE…
40321 14.529365873  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
40322 14.529405754  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#14] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161826 TSecr=3459749917 SLE…
40323 14.529570582  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40324 14.529605643  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#15] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161826 TSecr=3459749917 SLE…
40325 14.529979720  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
40326 14.529988032  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40327 14.530018204  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#16] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161826 TSecr=3459749917 SLE…
40328 14.530027213  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40329 14.530063392  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#17] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161826 TSecr=3459749917 SLE…
40330 14.530175978  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40331 14.530211039  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#18] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161826 TSecr=3459749917 SLE…
40332 14.530555573  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40333 14.530590634  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#19] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161827 TSecr=3459749917 SLE…
40334 14.530925180  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40335 14.530960381  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#20] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161827 TSecr=3459749917 SLE…
40336 14.531528900  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40337 14.531564030  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#21] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161828 TSecr=3459749917 SLE…
40338 14.531951308  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40339 14.531958082  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40340 14.531987766  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#22] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161828 TSecr=3459749917 SLE…
40341 14.532267485  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40342 14.532302756  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#23] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161828 TSecr=3459749917 SLE…
40343 14.532883218  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40344 14.532918349  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#24] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161829 TSecr=3459749917 SLE…
40345 14.533513198  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40346 14.533548818  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#25] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161830 TSecr=3459749917 SLE…
40347 14.533862202  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40348 14.533898800  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#26] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161830 TSecr=3459749917 SLE…
40349 14.534280629  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40350 14.534287823  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40351 14.534316319  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#27] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161830 TSecr=3459749917 SLE…
40352 14.534475281  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40353 14.534510132  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#28] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161831 TSecr=3459749917 SLE…
40354 14.534605328  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40355 14.534622020  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40356 14.534640738  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#29] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161831 TSecr=3459749917 SLE…
40357 14.534844608  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40358 14.534879600  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#30] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161831 TSecr=3459749917 SLE…
40359 14.535252280  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40360 14.535288598  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#31] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161831 TSecr=3459749917 SLE…
40361 14.535459154  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40362 14.535494354  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#32] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161832 TSecr=3459749917 SLE…
40363 14.535864520  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40364 14.535870597  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40365 14.535899511  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#33] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161832 TSecr=3459749917 SLE…
40366 14.536574051  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40367 14.536609322  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#34] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161833 TSecr=3459749917 SLE…
40368 14.536982631  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40369 14.536990383  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40370 14.537019647  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#35] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161833 TSecr=3459749917 SLE…
40371 14.537683432  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40372 14.537719331  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#36] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161834 TSecr=3459749917 SLE…
40373 14.537805866  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40374 14.537833454  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40375 14.537841905  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#37] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161834 TSecr=3459749917 SLE…
40376 14.538297139  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40377 14.538332340  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#38] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161834 TSecr=3459749917 SLE…
40378 14.538425160  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40379 14.538460082  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40380 14.538466996  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#39] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161835 TSecr=3459749917 SLE…
40381 14.538789320  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40382 14.538824381  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#40] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161835 TSecr=3459749917 SLE…
40383 14.539158298  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40384 14.539193429  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#41] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161835 TSecr=3459749917 SLE…
40385 14.539659699  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40386 14.539694620  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#42] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161836 TSecr=3459749917 SLE…
40387 14.539781923  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40388 14.539818381  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#43] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161836 TSecr=3459749917 SLE…
40389 14.539846737  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
40390 14.539885151  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#44] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161836 TSecr=3459749917 SLE…
40391 14.539905475  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40392 14.539940536  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#45] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161836 TSecr=3459749917 SLE…
40393 14.540278015  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40394 14.540314334  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#46] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161836 TSecr=3459749917 SLE…
40395 14.540561297  192.168.109.136  172.16.20.1      FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
40396 14.540568980  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40397 14.540598872  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#47] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161837 TSecr=3459749917 SLE…
40398 14.540630721  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
40399 14.540668645  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#48] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161837 TSecr=3459749917 SLE…
40400 14.540693649  192.168.109.136  172.16.20.1      FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
40401 14.540732132  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#49] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161837 TSecr=3459749917 SLE…
40402 14.540745752  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40403 14.540751828  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40404 14.540780813  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#50] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161837 TSecr=3459749917 SLE…
40405 14.541126184  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40406 14.541161106  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#51] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161837 TSecr=3459749917 SLE…
40407 14.541742057  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40408 14.541777676  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#52] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161838 TSecr=3459749917 SLE…
40409 14.542262594  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40410 14.542268670  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40411 14.542297725  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#53] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161838 TSecr=3459749917 SLE…
40412 14.542849481  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40413 14.542885031  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#54] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161839 TSecr=3459749917 SLE…
40414 14.543380146  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40415 14.543388387  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40416 14.543416254  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#55] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161840 TSecr=3459749917 SLE…
40417 14.543449779  192.168.109.136  172.16.20.1      FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40418 14.543457042  192.168.109.136  172.16.20.1      FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
40419 14.543485748  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#56] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161840 TSecr=3459749917 SLE…
40420 14.544326234  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40421 14.544361644  172.16.20.1      192.168.109.136  TCP        78 [TCP Dup ACK 40292#57] 38762 → 40235 [ACK] Seq=1 Ack=78368657 Win=1540608 Len=0 TSval=3403161841 TSecr=3459749917 SLE…
40422 14.544447411  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40423 14.544453557  192.168.109.136  172.16.20.1      FTP-DA…  2962 [TCP Fast Retransmission] FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
40424 14.544463614  172.16.20.1      192.168.109.136  TCP        78 38762 → 40235 [ACK] Seq=1 Ack=78371553 Win=1537792 Len=0 TSval=3403161841 TSecr=3459749948 SLE=78377345 SRE=78571377
40425 14.544467036  192.168.109.136  172.16.20.1      TCP      1514 [TCP Out-Of-Order] 40235 → 38762 [ACK] Seq=78371553 Ack=1 Win=65280 Len=1448 TSval=3459749949 TSecr=3403161817
40426 14.544482891  192.168.109.136  172.16.20.1      TCP      1514 [TCP Out-Of-Order] 40235 → 38762 [PSH, ACK] Seq=78373001 Ack=1 Win=65280 Len=1448 TSval=3459749949 TSecr=3403161817
40427 14.544487221  172.16.20.1      192.168.109.136  TCP        78 38762 → 40235 [ACK] Seq=1 Ack=78373001 Win=1536384 Len=0 TSval=3403161841 TSecr=3459749949 SLE=78377345 SRE=78571377
40428 14.544496999  172.16.20.1      192.168.109.136  TCP        78 38762 → 40235 [ACK] Seq=1 Ack=78374449 Win=1534976 Len=0 TSval=3403161841 TSecr=3459749949 SLE=78377345 SRE=78571377
40429 14.544575083  192.168.109.136  172.16.20.1      TCP      1514 [TCP Out-Of-Order] 40235 → 38762 [ACK] Seq=78374449 Ack=1 Win=65280 Len=1448 TSval=3459749949 TSecr=3403161817
40430 14.544583883  172.16.20.1      192.168.109.136  TCP        78 38762 → 40235 [ACK] Seq=1 Ack=78375897 Win=1533568 Len=0 TSval=3403161841 TSecr=3459749949 SLE=78377345 SRE=78571377
40431 14.545680901  192.168.109.136  172.16.20.1      TCP      1514 [TCP Retransmission] 40235 → 38762 [PSH, ACK] Seq=78375897 Ack=1 Win=65280 Len=1448 TSval=3459749949 TSecr=3403161817
40432 14.545699270  172.16.20.1      192.168.109.136  TCP        66 38762 → 40235 [ACK] Seq=1 Ack=78571377 Win=1407744 Len=0 TSval=3403161842 TSecr=3459749949
40433 14.546789234  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40434 14.547982356  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40435 14.547992832  172.16.20.1      192.168.109.136  TCP        66 38762 → 40235 [ACK] Seq=1 Ack=78574273 Win=1541632 Len=0 TSval=3403161844 TSecr=3459749949
40436 14.548474048  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
40437 14.548719405  192.168.109.136  172.16.20.1      FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
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45730 15.889149486  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87938489 Win=1541632 Len=0 TSval=3403163185 TSecr=3459751292
45731 15.889328981  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45732 15.889335407  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87941385 Win=1541632 Len=0 TSval=3403163186 TSecr=3459751293
45733 15.889480330  192.168.109.136      172.16.20.1          FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
45734 15.889495835  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87945729 Win=1541632 Len=0 TSval=3403163186 TSecr=3459751293
45735 15.889888909  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45736 15.889907697  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87951521 Win=1541632 Len=0 TSval=3403163186 TSecr=3459751293
45737 15.889946320  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45738 15.890067776  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45739 15.890073923  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87954417 Win=1541632 Len=0 TSval=3403163186 TSecr=3459751293
45740 15.890844636  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45741 15.890860630  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87957313 Win=1541632 Len=0 TSval=3403163187 TSecr=3459751295
45742 15.890944720  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45743 15.891445003  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45744 15.891451149  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87960209 Win=1541632 Len=0 TSval=3403163188 TSecr=3459751295
45745 15.891948638  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45746 15.891955134  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45747 15.891963585  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87963105 Win=1541632 Len=0 TSval=3403163188 TSecr=3459751295
45748 15.892055986  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45749 15.892103829  192.168.109.136      172.16.20.1          FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
45750 15.892119264  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87968897 Win=1541632 Len=0 TSval=3403163188 TSecr=3459751296
45751 15.893205945  192.168.109.136      172.16.20.1          FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
45752 15.893223546  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87976137 Win=1541632 Len=0 TSval=3403163189 TSecr=3459751297
45753 15.895644082  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45754 15.895663009  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87979033 Win=1541632 Len=0 TSval=3403163192 TSecr=3459751299
45755 15.895751150  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45756 15.895987637  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45757 15.895994272  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87981929 Win=1541632 Len=0 TSval=3403163192 TSecr=3459751299
45758 15.896282653  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45759 15.896292989  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45760 15.896317085  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87989169 Win=1541632 Len=0 TSval=3403163193 TSecr=3459751299
45761 15.896330844  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45762 15.896337968  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45763 15.896343276  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87992865 Win=1541632 Len=0 TSval=3403163193 TSecr=3459751300
45764 15.896766243  192.168.109.136      172.16.20.1          FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
45765 15.896784681  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87996409 Win=1541632 Len=0 TSval=3403163193 TSecr=3459751300
45766 15.897107075  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45767 15.897114618  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45768 15.897122859  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=87999305 Win=1541632 Len=0 TSval=3403163193 TSecr=3459751300
45769 15.897365702  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45770 15.897380928  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88002201 Win=1541632 Len=0 TSval=3403163194 TSecr=3459751301
45771 15.897989537  192.168.109.136      172.16.20.1          FTP-DA…  7306 FTP Data: 7240 bytes (PASV) (retr files/crab.mp4)
45772 15.898010699  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88009441 Win=1541632 Len=0 TSval=3403163194 TSecr=3459751302
45773 15.898030255  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45774 15.898045201  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88012337 Win=1541632 Len=0 TSval=3403163194 TSecr=3459751302
45775 15.899173369  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45776 15.899189293  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88015233 Win=1541632 Len=0 TSval=3403163195 TSecr=3459751303
45777 15.899276946  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45778 15.900056040  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45779 15.900062536  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88018129 Win=1541632 Len=0 TSval=3403163196 TSecr=3459751303
45780 15.900634547  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45781 15.901532653  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45782 15.901538799  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88021025 Win=1541632 Len=0 TSval=3403163198 TSecr=3459751303
45783 15.901742809  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45784 15.901819217  192.168.109.136      172.16.20.1          FTP-DA…  4410 FTP Data: 4344 bytes (PASV) (retr files/crab.mp4)
45785 15.901828157  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45786 15.901836468  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88026817 Win=1540736 Len=0 TSval=3403163198 TSecr=3459751305
45787 15.901842545  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88029713 Win=1541632 Len=0 TSval=3403163198 TSecr=3459751305
45788 15.901851205  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45789 15.901988935  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45790 15.901995081  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88032609 Win=1541632 Len=0 TSval=3403163198 TSecr=3459751305
45791 15.902677304  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45792 15.902686453  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45793 15.902708035  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88039849 Win=1541632 Len=0 TSval=3403163199 TSecr=3459751305
45794 15.902747915  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45795 15.902757064  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45796 15.902777458  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88047089 Win=1541632 Len=0 TSval=3403163199 TSecr=3459751306
45797 15.902779553  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45798 15.905186889  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45799 15.905193245  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88049985 Win=1541632 Len=0 TSval=3403163201 TSecr=3459751307
45800 15.905469542  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45801 15.905818825  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45802 15.905835098  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88054329 Win=1541632 Len=0 TSval=3403163202 TSecr=3459751309
45803 15.906084995  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45804 15.907069567  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45805 15.907079135  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45806 15.907089612  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88063017 Win=1541632 Len=0 TSval=3403163203 TSecr=3459751309
45807 15.907142971  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45808 15.907158686  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45809 15.907160502  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88068809 Win=1541632 Len=0 TSval=3403163203 TSecr=3459751310
45810 15.907315832  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45811 15.907325261  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45812 15.907333432  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88076049 Win=1541632 Len=0 TSval=3403163204 TSecr=3459751310
45813 15.907345375  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45814 15.907351941  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88078945 Win=1541632 Len=0 TSval=3403163204 TSecr=3459751311
45815 15.908299426  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45816 15.908661629  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45817 15.908667357  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88081841 Win=1541632 Len=0 TSval=3403163205 TSecr=3459751312
45818 15.908672385  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45819 15.909492896  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45820 15.909499182  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88084737 Win=1541632 Len=0 TSval=3403163206 TSecr=3459751313
45821 15.911051295  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45822 15.911071479  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88090529 Win=1541632 Len=0 TSval=3403163207 TSecr=3459751315
45823 15.911130217  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45824 15.911746089  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45825 15.911753143  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88093425 Win=1541632 Len=0 TSval=3403163208 TSecr=3459751315
45826 15.912032723  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45827 15.912042781  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45828 15.912051092  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88099217 Win=1541632 Len=0 TSval=3403163208 TSecr=3459751315
45829 15.912101030  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45830 15.912111227  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45831 15.912119957  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88106457 Win=1541632 Len=0 TSval=3403163208 TSecr=3459751316
45832 15.912128408  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45833 15.912135741  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88109353 Win=1541632 Len=0 TSval=3403163208 TSecr=3459751316
45834 15.912483418  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45835 15.913114237  192.168.109.136      172.16.20.1          FTP-DA…  2962 FTP Data: 2896 bytes (PASV) (retr files/crab.mp4)
45836 15.913129742  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88113697 Win=1541632 Len=0 TSval=3403163209 TSecr=3459751316
45837 15.914252113  192.168.109.136      172.16.20.1          FTP-DA…  5858 FTP Data: 5792 bytes (PASV) (retr files/crab.mp4)
45838 15.914270970  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88119489 Win=1541632 Len=0 TSval=3403163210 TSecr=3459751318
45839 15.914574716  192.168.109.136      172.16.20.1          FTP-DA…  1514 FTP Data: 1448 bytes (PASV) (retr files/crab.mp4)
45840 15.915318819  192.168.109.136      172.16.20.1          FTP-DA…  2314 FTP Data: 2248 bytes (PASV) (retr files/crab.mp4)
45841 15.915336489  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [ACK] Seq=1 Ack=88123186 Win=1541632 Len=0 TSval=3403163212 TSecr=3459751318
45842 15.940096935  192.168.109.136      172.16.20.1          FTP        90 Response: 226 Transfer complete.
45843 15.940915513  172.16.20.1          192.168.109.136      TCP        66 45878 → 21 [ACK] Seq=50 Ack=247 Win=64256 Len=0 TSval=3403163237 TSecr=3459751345
45844 15.940968384  172.16.20.1          192.168.109.136      FTP        72 Request: quit
45845 15.965379150  192.168.109.136      172.16.20.1          TCP        66 21 → 45878 [ACK] Seq=247 Ack=56 Win=65280 Len=0 TSval=3459751370 TSecr=3403163237
45846 15.966238493  192.168.109.136      172.16.20.1          FTP        80 Response: 221 Goodbye.
45847 15.966247224  192.168.109.136      172.16.20.1          TCP        66 21 → 45878 [FIN, ACK] Seq=261 Ack=56 Win=65280 Len=0 TSval=3459751370 TSecr=3403163237
45848 15.966274951  172.16.20.1          192.168.109.136      TCP        66 45878 → 21 [FIN, ACK] Seq=56 Ack=262 Win=64256 Len=0 TSval=3403163262 TSecr=3459751370
45849 15.966298526  172.16.20.1          192.168.109.136      TCP        66 38762 → 40235 [FIN, ACK] Seq=1 Ack=88123186 Win=1541632 Len=0 TSval=3403163262 TSecr=3459751318
45850 15.993307190  192.168.109.136      172.16.20.1          TCP        66 21 → 45878 [ACK] Seq=262 Ack=57 Win=65280 Len=0 TSval=3459751396 TSecr=3403163262
45851 15.993324022  192.168.109.136      172.16.20.1          TCP        66 40235 → 38762 [ACK] Seq=88123186 Ack=2 Win=65280 Len=0 TSval=3459751396 TSecr=3403163262
45852 16.024927112  Routerbo_2b:fa:06    Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
45853 16.566265427  172.16.20.1          172.16.2.1           DNS        86 Standard query 0xb278 PTR 26.114.82.140.in-addr.arpa
45854 16.583547146  172.16.2.1           172.16.20.1          DNS       131 Standard query response 0xb278 PTR 26.114.82.140.in-addr.arpa PTR lb-140-82-114-26-iad.github.com
45855 16.583709112  172.16.20.1          172.16.2.1           DNS        86 Standard query 0x55e5 PTR 41.163.237.44.in-addr.arpa
45856 16.599785836  172.16.2.1           172.16.20.1          DNS       149 Standard query response 0x55e5 PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
45857 17.142038530  140.82.114.26        172.16.20.1          TLSv1.2    91 Application Data
45858 17.142183454  172.16.20.1          140.82.114.26        TLSv1.2    95 Application Data
45859 17.288821677  140.82.114.26        172.16.20.1          TCP        66 443 → 48782 [ACK] Seq=26 Ack=30 Win=70 Len=0 TSval=3871456833 TSecr=304544409
45860 18.019770982  Routerbo_2b:fa:06    Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
45861 20.021841655  Routerbo_2b:fa:06    Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
45862 21.566257411  172.16.20.1          172.16.2.1           DNS        86 Standard query 0xff59 PTR 26.114.82.140.in-addr.arpa
45863 21.566961215  172.16.2.1           172.16.20.1          DNS       131 Standard query response 0xff59 PTR 26.114.82.140.in-addr.arpa PTR lb-140-82-114-26-iad.github.com
45864 21.567120666  172.16.20.1          172.16.2.1           DNS        86 Standard query 0x1d4f PTR 41.163.237.44.in-addr.arpa
45865 21.567675566  172.16.2.1           172.16.20.1          DNS       149 Standard query response 0x1d4f PTR 41.163.237.44.in-addr.arpa PTR ec2-44-237-163-41.us-west-2.compute.amazonaws.com
45866 22.023975024  Routerbo_2b:fa:06    Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
45867 24.026140219  Routerbo_2b:fa:06    Spanning-tree-(for-… STP        60 RST. Root = 32768/0/c4:ad:34:2b:fa:06  Cost = 0  Port = 0x8001
```


Segment Length (B)
Time (s)
0
1000
2000
3000
4000
5000
6000
7000
0
2
4
6
2·10⁷
4·10⁷
6·10⁷
8·10⁷


Segment Length (B)
Time (s)
Average Throughput (bit/s)
0
2000
4000
6000
8000
10000
0
2
4
6
8
2·10⁷
4·10⁷
6·10⁷
8·10⁷